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Glycolipopeptide biosurfactants

Abstract

Surfactants based on a newly discovered class of compounds include a hydrophobic lipid oligomer covalently linked to a peptide or peptide-like chain and a carbohydrate moiety, and a serine-leucinol dipeptide linked to the lipid oligomer. Such surfactants can be used to create an oil-in-water or water-in-oil emulsion by mixing together a polar component; a non-polar component; and the surfactant. Biosurfactants of the newly discovered class can be made by isolating and culturing a microorganism which produces the biosurfactant, and then isolating the biosurfactant from the culture. A microorganism can be engineered to produce biosurfactant of this newly discovered class by expressing a set of heterologous genes involved in the biosynthesis of the biosurfactant in the microorganism.

Inventors: Kerr; Russell Greig (Charlottetown, CA), Haltli; Bradley Arnold (New Haven, CA), Marchbank; Douglas Hubert (Stratford, CA), Berru ; Fabrice (Halifax, CA)

Applicant: Croda International Plc (Goole, GB)

Family ID: 58632356

Assignee: CRODA INTERNATIONAL PLC (Goole, GB)

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Primary Examiner: Saidha; Tekchand

Assistant Examiner: Meah; Mohammad Y

Attorney, Agent or Firm: Buchanan Ingersoll & Rooney PC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 16/817,193, filed Mar. 12, 2020, which is a Division of U.S. application Ser. No. 16/090,888, filed Oct. 3, 2018, now abandoned, which is the U.S. National Stage Application of International Application No. PCT/EP2017/058296, filed Apr. 6, 2017, which claims priority of GB Application No. 1605875.2, filed Apr. 6, 2016, the entirety of each of which are incorporated herein by reference for all purposes.

REFERENCE TO SEQUENCE LISTING

(1) The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled GLYCOLIPOPEPTIDE BIOSURFACTANTS created on Mar. 9, 2023 and which is 60 KB bytes in size. The information in the electronic format of the Sequence Listing is identical to that in International Application PCT/EP2017/058296, filed Apr. 6, 2017 and U.S. application Ser. No. 16/817,193, filed Mar. 12, 2020 and incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(2) This invention relates generally to the fields of surfactant chemistry, biochemistry, and microbiology. More specifically the invention relates to biosurfactants having a hydrophobic lipid oligomer covalently linked to a peptide or peptide-like (e.g. non-proteinogenic amino acid or single amino acid) chain and a carbohydrate moiety, various amino acid and nucleic acid sequences which encode components of biosynthetic pathways for these biosurfactants, and methods of making and using these biosurfactants.

BACKGROUND

(3) Surfactants are amphiphilic chemicals that possess both hydrophobic and hydrophilic moieties which allow them to interact with polar and non-polar systems. Surfactants exert their activity at interfaces between different phases (gas, liquid, solid) and as a result exhibit a range of functions including, but not limited to the ability to act as detergents, emulsifiers, wetting agents and foaming agents. Most chemical surfactants are alkyl sulfates or sulfonates derived from petro- or oleo-chemical sources. The use of these products has been steadily growing with an estimated worldwide consumption of 13 million tonnes in 2008 and an estimated market value of \$27 billion (USD) in 2012. In response to environmental and sustainability concerns, many companies utilizing chemical surfactants in their products have been exploring environmentally responsible alternatives as partial or full replacements for chemical surfactants. An alternative to chemical surfactants are biosurfactants, which are surface active molecules originating from microorganisms. These surfactants offer advantages over chemical surfactants such as production from sustainably produced feed stocks, biodegradability and lower toxicity.

SUMMARY

(4) It was discovered that the bacterium *Variovorax paradoxus* RKNM-096, deposited on Apr. 10, 2015 as accession number NRRL B-67038 under the terms of the Budapest Treaty with the Agricultural Research Service Culture Collection (NRRL, 1818 North University Street, Peoria, Illinois, 61064) produces a previously unknown class of biosurfactants termed “glycolipopeptides”. Unlike known biosurfactants, glycolipopeptides typically contain a hydrophobic lipid oligomer covalently linked to a peptide chain and a carbohydrate moiety. The deposit of NRRL B-67038 in support of this application was made by Nautilus Bioscience Canada Inc., 550 Univ. Ave., Charlottetown, PE, Canada, C1A4P3. Nautilus Bioscience Canada Inc. authorise the applicant to refer to the deposited biological material in this application and give their unreserved and irrevocable consent to the materials being made available to the public in accordance with appropriate national laws governing the deposit of these materials, such as Rule 31 and 33 EPC. The expert solution under Rule 32 EPC is also hereby requested.

(5) Described herein are purified biosurfactants that include a hydrophobic lipid component including a carboxyl end and a hydroxyl end, wherein the lipid component is covalently linked to (i) a peptide or peptide-like chain at the carboxyl end of the lipid component and (ii) a carbohydrate moiety at the hydroxyl end of the lipid component via a glycosidic linkage. The peptide or peptide-like chain can include a serine-leucinol dipeptide, the lipid component can include three β -hydroxyalkanoic acid moieties (e.g., wherein the length of each acyl chain of the lipid component is C.sub.6, C.sub.8, C.sub.10, or C.sub.12), and the carbohydrate moiety can include a rhamnose moiety attached to the lipid component via a glycosidic linkage. In certain embodiments, the carbohydrate moiety can include two rhamnose moieties and/or an acetyl group. Analogues and derivatives of these glycolipopeptides can be made by conventional methods.

(6) Glycolipopeptides can have the structure:

(7) ##STR00001##

wherein R.sub.1a is H, OH, OCH.sub.3, SH, S(CH.sub.3), NH.sub.2, NH(CH.sub.3), N(CH.sub.3).sub.2, or a peptide or peptide-like structure having the structure:

(8) ##STR00002##

wherein R.sub.1b, R.sub.1c, and R.sub.1d, are H, OH, OCH.sub.3, SH, S(CH.sub.3), NH.sub.2, NH(CH.sub.3), or N(CH.sub.3).sub.2; R.sub.2a, R.sub.2b, R.sub.2c, and R.sub.2d are each independently an amino acid side chain; X.sub.1a, X.sub.1b, X.sub.1c, and X.sub.1d are each independently one oxygen atom or two hydrogen atoms; X.sub.2a, X.sub.2b, X.sub.2c, and X.sub.2d are each independently NH, N(CH.sub.3), or O; R.sub.3a is a carbohydrate portion or a lipid monomer having the structure:

(9) ##STR00003##

or a lipid oligomer having the structure of:

(10) ##STR00004##

wherein X.sub.3a, X.sub.3b, X.sub.3c, and X.sub.3d are each independently NH, N(CH.sub.3), or O; R.sub.3a, R.sub.3b, R.sub.3c, and R.sub.3d includes a carbohydrate portion including a monomer having the structure:

(11) ##STR00005##

wherein R.sub.5a, R.sub.6a, R.sub.7a, and R.sub.8a are each independently a hydrogen atom, methyl, acetyl, or a carbohydrate; and R.sub.4a, R.sub.4b, R.sub.4c, and R.sub.4d are each independently a hydrogen atom, methyl, or a C.sub.2 to C.sub.19 saturated or unsaturated linear, branched-chain, cyclic, or aromatic hydrocarbon groups. Naturally occurring glycolipopeptides include those having the following structures:

(12) ##STR00006##

(13) Also described herein are emulsified compositions (e.g., oil-in-water or water-in-oil emulsions) including: a polar component, a non-polar component, and one or more of the above described biosurfactants; and a method of making an water-in-oil or oil-in-water emulsion by mixing together a polar component, a non-polar component, and one or more of the above described biosurfactants. Further described herein are a method of making one of the above described biosurfactants by (a) isolating a microorganism which includes the biosurfactant, (b) placing the microorganism in a culture under conditions that promote the synthesis of the biosurfactant, and (c) isolating the biosurfactant from the culture; and an isolated microorganism engineered to produce one of the above described biosurfactants, wherein a set of heterologous genes involved in the biosynthesis of the biosurfactant has been introduced into the microorganism.

(14) Unless otherwise defined, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Commonly understood definitions of chemical and biological terms can be found in Rieger et al., Glossary of Genetics: Classical and Molecular, 5th edition, Springer-Verlag: New York, 1991; and A Dictionary of Chemistry, Ed. J. Daintith, 7^{sup}.th Ed., Oxford University Press, 2016.

(15) As used herein, when referring to a chemical or molecule, the term “purified” means separated from components that occur with it in nature or in an artificially produced mixture. Typically, a molecule is purified when it is at least about 10% (e.g., at least 9%, 10%, 20%, 30% 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, 99.9%, and 100%), by weight (excluding solvent), free from components that occur with it in nature or in an artificially produced mixture. Purity can be measured by any appropriate method, e.g., column chromatography, polyacrylamide gel electrophoresis, or HPLC analysis.

(16) By “sequence identity” is meant the relatedness between two amino acid sequences or between two nucleotide sequences. Herein, the degree of identity between two amino acid sequences or two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, J. Mol. Biol. 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, Trends in Genetics 16: 276-277; <http://emboss.org>), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix for amino acid sequences or the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix for nucleotide sequence. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$(\text{Identical Amino Acid or Nucleotide Residues} \times 100) / (\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})$$

(17) Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. All patents, patent applications, and publications mentioned herein are incorporated by reference in their entirety. In the case of conflict, the present specification, including definitions will control. In

addition, the particular embodiments discussed below are illustrative only and not intended to be limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of selected HMBC (.sup.1H.fwdarw..sup.13C) and COSY correlations (bold bonds) of NB-RLP1006 and assigned fragment ions from MS/MS collision-induced dissociation of the glycolipopeptides.
- (2) FIG. 2 is a denaturing polyacrylamide gel showing purified His-tagged RlpE (A) and the UPLC-HRMS analysis of enzyme reactions in which the enzyme was incubated with NB-RLP860 and dTDP-L-rhamnose.
- (3) FIG. 3 is a schematic comparison of the *V. paradoxus* RKNM-096 glycolipopeptide gene cluster to homologous gene clusters identified in *I. limosus* DSM 16000 and *J. agaricidamnorum* DSM 9628. Genes encoding proteins homologous to proteins in the *V. paradoxus* gene cluster are indicated by arrow filling patterns. Identity and similarity to *V. paradoxus* proteins is indicated under arrows (identity %/similarity %). NRPS domain organization is indicated under arrows representing genes encoding non-ribosomal peptide synthetases (NRPSs). Domains: C—condensation, A—adenylation, T—thiolation/peptidyl-carrier protein, R—reductase. Subscript notation indicates putative A-domain substrate. Labels above arrows in the *I. limosus* and *J. agaricidamnorum* gene clusters indicate protein IDs.
- (4) FIG. 4 is the nucleic acid sequence SEQ ID NO:1.
- (5) FIG. 5 is the nucleic acid sequence SEQ ID NO:2.
- (6) FIG. 6 is the nucleic acid sequence SEQ ID NO:3.
- (7) FIG. 7 is the nucleic acid sequence SEQ ID NO:5.
- (8) FIG. 8 is the nucleic acid sequence SEQ ID NO:7.
- (9) FIG. 9 is the nucleic acid sequence SEQ ID NO:9.
- (10) FIG. 10 is the nucleic acid sequence SEQ ID NO:11.
- (11) FIG. 11 is the nucleic acid sequence SEQ ID NO:13.
- (12) FIG. 12 is the nucleic acid sequence SEQ ID NO:15.
- (13) FIG. 13 is the nucleic acid sequence SEQ ID NO:17.
- (14) FIG. 14 is the nucleic acid sequence SEQ ID NO:19.
- (15) FIG. 15 is the nucleic acid sequence SEQ ID NO:21.
- (16) FIG. 16 is the amino acid sequence SEQ ID NO:4.
- (17) FIG. 17 is the amino acid sequence SEQ ID NO:6.
- (18) FIG. 18 is the amino acid sequence SEQ ID NO:8.
- (19) FIG. 19 is the amino acid sequence SEQ ID NO:10.
- (20) FIG. 20 is the amino acid sequence SEQ ID NO:12.
- (21) FIG. 21 is the amino acid sequence SEQ ID NO:14.
- (22) FIG. 22 is the amino acid sequence SEQ ID NO:16.
- (23) FIG. 23 is the amino acid sequence SEQ ID NO:18.
- (24) FIG. 24 is the amino acid sequence SEQ ID NO:20.
- (25) FIG. 25 is the amino acid sequence SEQ ID NO:22.
- (26) FIG. 26 is the amino acid sequence SEQ ID NO:23.
- (27) FIG. 27 is the amino acid sequence SEQ ID NO:24.

DETAILED DESCRIPTION

- (28) The invention encompasses glycolipopeptide surfactant compositions, methods of making and using such biosurfactants, and bacteria and bacterial culture that produce glycolipopeptides. The below described preferred embodiments illustrate adaptation of these compositions and methods.

Nonetheless, from the description of these embodiments, other aspects of the invention can be made and/or practiced based on the description provided below.

General Methodology

(29) Methods involving conventional organic chemistry, biochemistry, microbiology, and molecular biology are described herein. Such methods are described in, e.g., Clayden et al., Organic Chemistry, Oxford University Press, 1st edition (2000); Molecular Cloning: A Laboratory Manual, 2nd ed., vol. 1-3, Sambrook et al., ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 2001; Current Protocols in Molecular Biology, Ausubel et al., ed., Greene Publishing and Wiley-Interscience, New York; and in the various volumes of Methods in Microbiology and Methods in Biochemistry and Molecular Biology both published by Elsevier.

Glycolipopeptides

(30) Naturally occurring glycolipopeptides and synthetic analogues and derivatives thereof typically include a hydrophobic lipid component including a carboxyl end and a hydroxyl end, wherein the lipid component is covalently linked to (i) a peptide or peptide-like chain at the carboxyl end of the lipid component and (ii) a carbohydrate moiety at the hydroxyl end of the lipid component via a glycosidic linkage.

(31) The peptide chain may comprise in the range of between 2 and 10 amino acids, preferably 2 to 8, more preferably 2 to 4 amino acids. The peptide chain may most preferably comprise 2 amino acids. The peptide or peptide-like chain can comprise and/or consist of a serine-leucinol dipeptide.

(32) The lipid component may comprise in the range of between 1 and 6 alkanoic acid moieties, preferably 2 to 4, and more preferably 3. Most preferably the lipid component can include three β -hydroxyalkanoic acid moieties. The length of each acyl chain of the lipid component may be in the range of between C.sub.4 to C.sub.20, preferably C.sub.6 to C.sub.16, more preferably C.sub.8 to C.sub.14. Most preferably the length of each acyl chain may be selected from C.sub.8, C.sub.10, or C.sub.12.

(33) The carbohydrate moiety may be selected from saccharides including glucose, fructose, galactose, mannose, ribose, or deoxy saccharide variants including deoxyribose, fucose, or rhamnose. Preferably the carbohydrate moiety is rhamnose. In particular, a rhamnose moiety attached to the lipid component via a glycosidic linkage. In certain embodiments, the carbohydrate moiety can include one, two, or three rhamnose moieties and/or an acetyl groups. Preferably the carbohydrate moiety includes two.

(34) Glycolipopeptides can include the structure:

(35) ##STR00007##

wherein R.sub.1a is H, OH, OCH.sub.3, SH, S(CH.sub.3), NH.sub.2, NH(CH.sub.3), N(CH.sub.3).sub.2, or a peptide or peptide-like structure having the structure:

(36) ##STR00008##

wherein R.sub.1b, R.sub.1c, and R.sub.1d, are H, OH, OCH.sub.3, SH, S(CH.sub.3), NH.sub.2, NH(CH.sub.3), or N(CH.sub.3).sub.2; R.sub.2a, R.sub.2b, R.sub.2c, and R.sub.2d are each independently an amino acid side chain; X.sub.1a, X.sub.1b, X.sub.1c, and X.sub.1d are each independently one oxygen atom or two hydrogen atoms; X.sub.2a, X.sub.2b, X.sub.2c, and X.sub.2d are each independently NH, N(CH.sub.3), or O; R.sub.3a is a carbohydrate portion or a lipid monomer having the structure:

(37) ##STR00009##

or a lipid oligomer having the structure of:

(38) ##STR00010##

wherein X.sub.3a, X.sub.3b, X.sub.3c, and X.sub.3d are each independently NH, N(CH.sub.3), or O; R.sub.3a, R.sub.3b, R.sub.3c, and R.sub.3d includes a carbohydrate portion including a monomer having the structure:

(39) ##STR00011##

(40) wherein R.sub.5a, R.sub.6a, R.sub.7a, and R.sub.8a are each independently a hydrogen atom,

methyl, acetyl, or a carbohydrate; and R.sub.4a, R.sub.4b, R.sub.4c, and R.sub.4d are each independently a hydrogen atom, methyl, or a C.sub.2 to C.sub.19 saturated or unsaturated linear, branched-chain, cyclic, or aromatic hydrocarbon groups. In the foregoing, at least one of R.sub.6a, R.sub.7a, and R.sub.8a can include a carbohydrate monomer having the structure:

(41) ##STR00012##

wherein R.sub.5b, R.sub.6b, R.sub.7b, and R.sub.8b are each independently a hydrogen atom, methyl, acetyl, or a carbohydrate.

(42) In certain embodiments the peptide or peptide-like portion includes at least one proline or proline-like monomer having the structure:

(43) ##STR00013##

wherein X.sub.4 is one oxygen atom or two hydrogen atoms, or a single proline or proline-like monomer or a terminal proline or proline-like monomer having the structure:

(44) ##STR00014##

wherein R.sub.9 is of H, OH, OCH.sub.3, SH, S(CH.sub.3), NH.sub.2, NH(CH.sub.3), or N(CH.sub.3).sub.2; and X.sub.4 is one oxygen atom or two hydrogen atoms.

(45) Glycolipopeptides can have the following structures:

(46) ##STR00015##

wherein R.sub.5a, R.sub.6a, R.sub.7a, R.sub.10, and R.sub.11 are each independently a hydrogen atom or acetyl; and n.sub.1, n.sub.2, and n.sub.3 are integers each independently ranging from 1 to 7;

(47) ##STR00016##

wherein R.sub.5a, R.sub.5b, R.sub.6b, R.sub.7a, R.sub.7b, R.sub.10, and R.sub.11 are each independently a hydrogen atom or acetyl; and n.sub.1, n.sub.2, and n.sub.3 are integers each independently ranging from 1 to 7;

(48) ##STR00017##

Derivatives, analogues, and other variants of the foregoing glycolipopeptides can be made by one of skill in the art. For instance, the amino acid composition and length of the peptide chain could be modified in a combinatorial fashion, introducing either proteinogenic or unnatural amino acids to modulate the solubility, hydrophilic-lipophilic balance (HLB), and other surfactant characteristics of the glycolipopeptides. The peptide portion may also contain amino acids with charged functional groups, which may result in cationic, anionic, or zwitterionic surfactants with unique surfactant applications. The carboxylic acid functionality at the C-terminus position of the peptide may also be reduced to a primary hydroxyl group. Similarly, the lipid portion may contain various numbers (e.g., 1, 2, 3, 4 or more) of β -hydroxyalkanoate units, which themselves may be comprised of C.sub.2 to C.sub.19 saturated or unsaturated linear, branched-chain, cyclic, or aromatic hydrocarbon groups. The rhamnose moieties could be linked together via 1,2-, 1,3-, or 1,4-glycosidic linkages, which may possess either the α - or β -configuration. In addition to rhamnose, the carbohydrate portion may also be composed of glucose or other monosaccharide units.

(49) Variants of the *Variovorax paradoxus* RKNM-096 glycolipopeptide biosurfactants that have altered properties could be made. Altered properties of such variants may include, but are not limited to, alterations in emulsification, foaming and surface tension reducing properties exhibited under differing physiochemical conditions such as, but not limited to, temperature, pH, and salinity.

(50) The variovaricins describe herein may be at least 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 95, 99, 99.5, 99.9, or 99.99 percent purified (by weight). They may be in crystalline or non-crystalline (amorphous) form, and in some cases also be obtained as salts derived from such organic and inorganic acids as: acetic, trifluoroacetic, lactic, citric, tartaric, formate, succinic maleic, malonic, gluconic, hydrochloric, hydrobromic, phosphoric, nitric, sulfuric, methane sulfonic and similarly known acids. The salts can be prepared by adapting commonly known procedures.

(51) In some embodiments, the composition includes additional compounds such as carriers, other surfactants (e.g., non-glycolipopeptide surfactants), or biologically active compounds (non-

glycolipopeptide surfactants, such as pharmaceutical agents or other non-glycolipopeptide antimicrobial agents). The addition of the aforementioned agents to glycolipopeptide surfactants can be selected by one skilled in the art based on the chosen application.

(52) The composition can include a carrier, such as conventional pharmaceutically acceptable carriers as described in *Remington: The Science and Practice of Pharmacy*, The University of the Sciences in Philadelphia, Editors, Lippincott, Williams, & Wilkins, Philadelphia, Pa., 21st Edition (2005). Pharmaceutically acceptable carriers vary depending on the mode of administration. Fluid formulations used for parenteral injection may include fluids such as water, physiological saline, aqueous dextrose or glycerol. Solid formulations may include highly purified solid carriers such as magnesium stearate, starch, or lactose. Pharmaceutical compositions may also contain minor quantities of non-toxic auxiliary substances, such as buffers and preservatives.

(53) In some embodiments, the compositions include a non-glycolipopeptide surfactant. Examples include non-ionic, cationic, anionic and amphoteric surfactants. Representative examples of anionic surfactants include carboxylates, sulfonates, petroleum sulfonates, alkylbenzene sulfonates, naphthalene sulfonates, olefin sulfonates, alkyl sulfates, sulfates, sulfated natural oils and fats, sulfated esters, sulfated alkanolamides, alkylphenols, ethoxylated and sulfated alkylphenols and rhamnolipids. Examples of cationic surfactants include quaternary ammonium salts, N, N, N', N' tetrakis substituted ethylenediamines and 2-alkyl-1-hydroxyethyl-2-imidazolines. Examples of non-ionic surfactants include ethoxylated aliphatic alcohols, polyoxyethylene surfactants, carboxylic esters, polyethylene glycol esters, anhydrosorbitol ester and ethoxylated derivatives, glycol esters of fatty acids, carboxylic amides, monoalkanolamine condensates and polyoxyethylene fatty acid amides. Examples of amphoteric surfactants include sodium salts of N-coco-3-aminopropionic acid, N-tallow-3-iminodipropionate and N-cocoamidethyl-N-hydroxyethylglycine, as well as N-carboxymethyl-N-dimethyl-N-(9-octadecenyl) ammonium hydroxide. In further embodiments, the composition includes one or more food or food additive, cosmetic or pharmaceutical agents or antimicrobial agents (such as an antibacterial or antifungal agents).

Methods of Making Glycolipopeptides

(54) The glycolipopeptides described herein may be made by isolation or purification from bacteria strains which produce them, such as *Variovorax paradoxus* RKNM-096. As described in the Examples section below, bacteria which produce one or more glycolipopeptides can be isolated from natural habitats or obtained from publicly accessible sources. Bacteria can be determined to produce glycolipopeptides by the methods described in the Examples. The glycolipopeptide-producing bacterium can be placed in a bioreactor (vessel) containing suitable culture medium, and then incubated under conditions that promote bacterial replication and production of one or more glycolipopeptides. The produced glycolipopeptide(s) can be purified or isolated from the culture mixture by conventional techniques such as extraction followed by chromatographic separation (e.g., using ultra high performance liquid chromatography). Chemical analyses (determination of molecular weight, melting point, NMR, IS spectroscopy, etc.) can be performed to confirm the structure and purity of the isolated glycolipopeptide(s). Alternatively, the glycolipopeptides described herein may be made by total synthesis or semi-synthesis, e.g. as described herein.

Glycolipopeptides Gene Clusters and Methods of Use

(55) As described in Example 7 below, the glycolipopeptide and rhamnose biosynthetic gene clusters of *V. paradoxus* RKNM-096 were characterized. The polypeptides encoded in the gene cluster function in a coordinated fashion to synthesize the NB-RLP series of biosurfactants. The nucleotide sequence encoding these genes and the amino acid sequences of the corresponding polypeptides are shown in the sequence listing. Other amino acid sequences and the nucleic acid sequences that share at least 70% (e.g., at least 70, 80, 90, 95, 97, 98, or 99%) sequence identity with those shown in the sequence listing might also be used in the methods and compositions described herein particularly when such other sequences exhibit (or encode a molecule exhibiting) at least 50% (e.g., at least 50, 60, 70, 80, 90, or 100%) of the corresponding native polypeptide

enzymatic activity. Nucleic acid sequences which encode the same polypeptides described herein but are not included in the sequence listing might also be used.

(56) The foregoing polynucleotides might be used in a method for producing recombinant biosynthetic enzymes. As one example, such a method might include culturing a host cell (e.g., *E. coli* or another suitable prokaryotic or eukaryotic host cell) which contains an expression vector having a nucleic acid sequence of one or more of SEQ ID NO:1, SEQ ID NO:2, SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19 and SEQ ID NO:21 in a culture medium under conditions suitable for expression of the recombinant protein in the host cell, and b) isolating the recombinant protein(s) from the host cell or the culture medium.

(57) Also contemplated is method of producing a glycolipopeptide in a heterologous host cell by expressing the complete or partial biosynthetic gene cluster. This method might include the steps of a) culturing a host cell which contains an expression vector having nucleic acid sequences comprising SEQ ID NO:1, SEQ ID NO:2, SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19 and SEQ ID NO:21 in a culture medium under conditions suitable for expression of the recombinant proteins in the host cell, and b) isolating produced glycolipopeptides from the culture medium.

(58) Further contemplated are methods for using a nucleic acid molecule that hybridizes to or includes a portion of SEQ ID NO:1, SEQ ID NO:2, SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19 or SEQ ID NO:21 as a probe or PCR primer to identify other organisms capable of producing glycolipopeptides or structurally similar biosurfactants.

Synthesis

(59) The compounds of the present invention may be achieved using chemical methods as noted herein.

(60) The total synthesis of the glycolipopeptides can be achieved using established synthetic methodology to assemble commercially available building blocks. A retrosynthetic analysis of NB-RLP1006 (1) demonstrates the feasibility of the total synthesis. As an example, one skilled in the art of organic synthesis may couple the dipeptide substituent (4) to the tridecanoic acid (5) and perform a chemical glycosylation of the lipopeptide intermediate (2) using glycosyl donor 3 as shown below. The dipeptide moiety can be prepared using standard amide coupling methods, while the tridecanoic acid can be generated from commercially available ethyl trans-2-decenoate.

Meanwhile, the α -1,3-linked dirhamnose substituent (3) can be assembled using glycosyl donor 6 and glycosyl acceptor 7. It is understood that this general approach, or other similar approaches in which one assembles commercially available starting materials, could enable the synthesis of glycolipopeptide analogues. For instance, different amino acids can be incorporated into the peptide or peptide-like portion while the length of the peptide chain can be increased or decreased. Similarly, structural modifications could be made to the lipid and carbohydrate portions of the glycolipopeptides to produce analogues with potentially useful biosurfactant characteristics.

(61) To generate the carbohydrate substituent, α -1,3 linked dirhamnose, a number of protecting group manipulations must be performed to enable regioselective glycosylation of the rhamnose sugar at the 3-OH position (Scheme 2). The p-methoxyphenyl α -L-rhamnopyranoside (8), which serves as a synthetic precursor to both rhamnose moieties, can be synthesized from commercially available L-rhamnose in three steps. The terminal rhamnose sugar can then be prepared by perbenzylation of 8 and removal of the p-methoxyphenyl substituent to allow synthesis of the rhamnosyl trichloroacetimidate (9). Meanwhile, a six-step sequence of protecting group manipulations can provide p-methoxyphenyl 2,4-di-O-benzyloxyrhamnopyranoside (10) (Cai, X.; et al. Carbohydr. Res. 2010, 345, 1230), which can be glycosylated at the 3-OH position to achieve the α -1,3 glycosidic linkage between the two rhamnose substituents. The anomeric effect is expected to direct the formation of an α -glycosidic linkage with high stereoselectivity in this

chemical glycosylation (Takahashi, O.; et al. Carbohydr. Res. 2007, 342, 1202).

(62) ##STR00018##

(63) To assemble the dirhamnose substituent, glycosyl donor 9 can be linked to glycosyl acceptor 10 through activation of the anomeric trichloroacetimidate using either BF₃·OEt₂ or TMSOTf (Scheme 3). The anomeric p-methoxyphenyl protecting group must then be replaced with a good leaving group, such as a trichloroacetimidate, to enable glycosylation of the decanoic acid moiety. Alternatively, a thiophenyl group could be installed instead of the p-methoxyphenyl group during reaction A of Scheme 2. This approach would allow an orthogonal glycosylation to be pursued given the dual role of the anomeric thiophenyl group as a protecting and leaving group (Gampe, C. M.; et al. Tetrahedron 2011, 67, 9771; Wu, C.-Y.; Wong, C.-H. Top. Curr. Chem. 2011, 301, 223). It is known that the total synthesis of NB-RLP860 could be achieved using rhamnosyl donor 9 or other suitable rhamnosyl donors. Furthermore, it is recognized that a skilled chemist could modify the carbohydrate moiety of the glycolipopeptides by using glycosyl donors other than 9, 11, or 12.

(64) ##STR00019## ##STR00020##

(65) To assemble the tridecanoic acid moiety, commercially available ethyl trans-2-decenoate (13) can serve as a precursor to the synthesis of 3-hydroxydecanoic acid (14) through an established five-step β -oxidation (Schneekloth, J. S.; et al. Bioorg. Med. Chem. Lett. 2006, 16, 3855; Pandey, S. K.; Kumar, P. Eur. J. Org. Chem. 2007, 369). Overall yields reported in the literature for this synthesis vary between 50-85% (Scheme 4). Although ($\pm$)-3-hydroxydecanoic acid is also commercially available, this racemic precursor is cost-prohibitive and does not likely represent an economically viable route. Upon generating the 3-(tert-butyldimethylsilyl) decanoic acid (15), this building block can be linked twice to 14 via Steglich esterification to generate the silylated di- and tri-decanoic acid (16 and 17, respectively) of NB-RLP1006 and other glycolipopeptides. In this approach, carboxylic acid 15 is activated as a N-hydroxysuccinimide ester prior to the addition of 14, obviating an additional protecting group for the carboxylic acid functionality of 14.

(66) ##STR00021##

(67) An alternative approach is also available in which the carboxylic acid group of 14 is protected as a benzyl ester before esterification (Scheme 5). In this approach, building blocks 15 and 18 are linked together in a synthesis that requires additional steps for installing and removing silyl ether and benzyl ester protecting groups. It is known that a chemist skilled in the art of organic synthesis could utilize either approach to introduce C₂ to C₁₉ saturated or unsaturated linear, branched-chain, cyclic, or aromatic hydrocarbon moieties in order to modify the lipid portion of the glycolipopeptides. It is anticipated that analogues generated through this approach may also exhibit surfactant properties.

(68) ##STR00022##

(69) The leucinol-serine dipeptide can be assembled from commercially available Boc-leucinol (19) and Fmoc-Ser(Bzl)-OH (20) using well-established amide coupling chemistry (Scheme 6) (Valeur, E.; Bradley, M. Chem. Soc. Rev. 2009, 38, 606). The five-step reaction sequence involves protecting the primary hydroxyl group of 19 as a benzyl ether and coupling the two amino acids before removing the 9-fluorenylmethyloxycarbonyl (Fmoc) protecting group to generate the leucinol-serine dipeptide (21) that is poised for amide coupling to the decanoic acid. It is conceivable that other commercially available amino acids, including but not limited to D-amino acids and β -amino acids, could be assembled in a similar fashion to introduce structural modifications at the peptide portion of the glycolipopeptide. The C-terminus of the peptide or peptide-like portion could exist as a carboxylic acid functionality or be reduced to a primary hydroxyl group. Other modifications of the C-terminus position include, but are not limited to, alkylation, acylation, glycosylation, phosphorylation, and sulfation. The chain length of the peptide or peptide-like portion could be increased by coupling additional amino acid monomers to the dipeptide intermediate. Alternatively, a single amino acid monomer could be coupled to the

tridecanoic acid intermediate (17) to decrease the chain length.

(70) ##STR00023##

(71) The tridecanoic acid (17) can readily undergo amide coupling to the benzylated dipeptide (21), upon which the tert-butyldimethyl silyl ether protecting group can be removed using tetrabutylammonium fluoride to provide glycosyl acceptor 22 (Scheme 7). Glycosylation of 22 with either 11 or 12, followed by global deprotection via hydrogenolysis of the benzyl ethers, furnishes the deprotected glycolipopeptide NB-RLP1006.

(72) ##STR00024##

(73) Although the total synthesis of NB-RLP1006 may require between 14-18 steps (longest linear sequence, 34-40 steps total), the synthesis could be expedited by utilizing solid-phase synthetic techniques in which the terminal leucinol residue is immobilized onto a solid support. For example, the Leucinol(Bzl) (19) can be tethered to a polystyrene-bound p-alkoxybenzyl hydroxyl group (Wang resin) through a silyl ether linkage (Scheme 8) (Scott, P. J. H. Linker Strategies in Solid-Phase Synthesis, John Wiley & Sons Ltd: Chichester, U.K., 2009; pp 50-51). Following previously described amide coupling and Steglich esterification methodologies (Coin, I.; et al. Nat. Protoc. 2007, 2, 3247.), the remaining serine and decanoic acid residues can then be attached in a step-wise approach using Fmoc-Ser(Bzl)-OH (20) and 3-(tert-butyldimethylsilyl)decanoic acid (23). After releasing the lipopeptide intermediate, the primary hydroxyl group can be selectively protected as a tert-butyldiphenylsilyl ether to provide glycosyl acceptor 24. The glycolipopeptide NB-RLP1006 can then be synthesized by chemical glycosylation and removal of the silyl and benzyl ether protecting groups. Analogues of the glycolipopeptides can also be produced using solid-phase synthetic techniques as described in the foregoing solution-phase synthesis of NB-RLP1006.

(74) ##STR00025##

Semisynthesis of the Glycolipopeptides

(75) Synthetic analogues of NB-RLP1006 and other glycolipopeptides may also be of interest for assessing the structure-activity relationships of this class of biosurfactants. Unlike the total synthesis, a semisynthesis could represent a rapid approach for developing a number of glycolipopeptide analogues. For instance, strategies may involve a semisynthesis of the tridecanoic acid (23) by acid hydrolysis of the glycolipopeptide mixture (Scheme 9). See Miao, S.; et al. J. Agric. Food Chem. 2015, 63, 3367. Tridecanoic acid (23) could then be coupled to the peptide portion and glycosylated with commercially available disaccharides, such as lactose or maltose, to generate novel glycolipopeptide analogues (e.g. 24). The aglycone of glycolipopeptides may also be produced by *V. paradoxus* RKNM-096 and undergo chemical glycosylation to produce similar analogues. It is also known that the rhamnolipids could be utilized as an advanced precursor and linked to various dipeptides (e.g. 21) through the carboxylic acid functional group to produce glycolipopeptides similar to NB-RLP1006 (Scheme 10). Given the commercial availability of the rhamnolipids, conceivably one skilled in the art of organic synthesis would also recognize that peptide chains other than leucinol-serine could be introduced to expand on the structural diversity of glycolipopeptide analogues accessible through this semisynthetic approach.

(76) ##STR00026##

(77) ##STR00027##

(78) Conceivably one skilled in the art of organic synthesis could isolate naturally occurring glycolipopeptides from a microbial fermentation and synthesize derivatives, analogues, and other structural variants. For instance, modifications that could occur at R.sub.5a, R.sub.5b, R.sub.6a, R.sub.6b, R.sub.7a, R.sub.7b, R.sub.10, and R.sub.11 include, but are not limited to, alkylation, acylation, glycosylation, phosphorylation, and sulfation. The glycolipopeptides could also undergo a base hydrolysis to produce rhamnolipid-like compounds with potentially useful surfactant properties. It is also known that a base hydrolysis reaction would provide NB-RLP374.

Methods of Use

(79) The glycolipopeptides described herein might be used similarly to other surfactants. They may,

for example, be used as detergents, emulsifiers, dispersants, wetting agents, foaming agents, or biofilm inhibitors/disruptors. A typical use would be for the preparation of emulsions for cosmetic or pharmaceutical formulations (eg., water-in-oil or oil-in-water emulsions), where one or more glycolipopeptides or derivatives or analogues thereof is mixed with a polar component and a non-polar component.

(80) The properties of the surfactants of this invention also make them suitable as emulsifiers particularly in oil in water or water-in-oil emulsions e.g. in personal care applications. Personal care emulsion products can take the form of creams and milks desirably and typically include emulsifier to aid formation and stability of the emulsion. Typically, personal care emulsion products use emulsifiers (including emulsion stabilisers) in amounts of about 3 to about 5% by weight of the emulsion.

(81) The oil phase of such emulsions are typically emollient oils of the type used in personal care or cosmetic products, which are oily materials which is liquid at ambient temperature or solid at ambient temperature, in bulk usually being a waxy solid, provided it is liquid at an elevated temperature, typically up to 100° C. more usually about 80° C., so such solid emollients desirably have melting temperatures less than 100° C., and usually less than 70° C., at which it can be included in and emulsified in the composition.

(82) The concentration of the oil phase may vary widely and the amount of oil is typically from 1 to 90%, usually 3 to 60%, more usually 5 to 40%, particularly 8 to 20%, and especially 10 to 15% by weight of the total emulsion. The amount of water (or polyol, e.g. glycerin) present in the emulsion is typically greater than 5%, usually from 30 to 90%, more usually 50 to 90%, particularly 70 to 85%, and especially 75 to 80% by weight of the total composition. The amount of surfactant used in such emulsions may be in the range from 0.001 to 10% by weight of the emulsion, preferably 0.01 to 6% by weight, more preferably 0.1 to 5% by weight, further preferably 1 to 3% by weight. The amount of surfactant used on such emulsions is typically from 2 to 5.5%, by weight of the emulsion.

(83) The end uses formulations of such emulsions include moisturizers, sunscreens, after sun products, body butters, gel creams, high perfume containing products, perfume creams, baby care products, hair conditioners, skin toning and skin whitening products, water-free products, anti-perspirant and deodorant products, tanning products, cleansers, 2-in-1 foaming emulsions, multiple emulsions, preservative free products, emulsifier free products, mild formulations, scrub formulations e.g. containing solid beads, silicone in water formulations, pigment containing products, sprayable emulsions, colour cosmetics, conditioners, shower products, foaming emulsions, make-up remover, eye make-up remover, and wipes, A preferred formulation type is a sunscreen containing one or more organic sunscreens and/or inorganic sunscreens such as metal oxides, but desirably includes at least one particulate titanium dioxide and/or zinc oxide.

(84) All of the features described herein may be combined with any of the above aspects, in any combination. It is to be understood that the invention is not to be limited to the details of the above embodiments, which are described by way of example only. Many variations are possible.

(85) In order that the present invention may be more readily understood, reference will now be made, by way of example, to the following description.

EXAMPLES

Example 1: Isolation of *Variovorax paradoxus* RKNM-096.

(86) Bacterial strain RKNM-096 was isolated from soil collected from the Battle Bluffs area west of Kamloops, British Columbia. RKNM-096 was isolated as a mucoid, yellow pigmented colony, and purified by serial subculturing. The bacterium was identified by 16S rRNA gene analysis, which indicated that RKNM-096 was a strain of *V. paradoxus*.

Example 2: Identifying *Variovorax paradoxus* RKNM-096 as a Biosurfactant Producer

(87) *V. paradoxus* RKNM-096 was identified as a biosurfactant producer in a screen aimed at identifying bacterial producers of biosurfactants with emulsifying properties. The assay utilized to

identify bacterial producers of biosurfactants was the emulsification activity assay. In this assay cultures were grown in 10 mL of liquid medium in 25 mm×150 mm glass tubes at 30° C. with shaking at 200 rpm for 5 days. After 5 days, the cells were removed by centrifugation and 3.5 mL of cell free culture broth was mixed with 3.5 mL of kerosene in a 13 mm×100 mm test tube with a screw cap tube. The tubes were vortexed for two minutes and then allowed to stand overnight at room temperature after which the height of the emulsion (h.sub.emuls) and the total height (h.sub.total) of the liquid in the tube were measured. The emulsification index (E.sub.24) was calculated using the equation $E_{sub.24} = h_{sub.emuls} / h_{sub.total} \times 100\%$. Fermentation broths of *V. paradoxus* RKNM-096 cultured in ISP2 broth (0.4% maltose, 0.4% yeast extract, 1.0% dextrose, pH 7.0) exhibited an E.sub.24 value of 50.7%.

Example 3: Identification of Glycolipopeptide Biosurfactants Produced by *Variovorax paradoxus* RKNM-096

(88) To determine if a small molecule was responsible for the observed emulsification activity, *V. paradoxus* RKNM-096 was fermented in ISP2 broth as described above and the broth was extracted twice with 10 mL of ethyl acetate (EtOAc). The EtOAc extract was then washed twice with 10 mL of water to remove any remaining polar media components from the EtOAc extract. For comparison purposes an ISP2 media blank was extracted in an identical manner. The EtOAc extracts were evaporated in vacuo and reconstituted in CH.sub.3OH at a concentration of 0.5 mg/mL.

(89) The extracts were separated by ultra high performance liquid chromatography (UPLC; Accela™, Thermo Fisher Scientific Mississauga, ON, Canada) and the eluates analyzed with a photodiode array detector (200-600 nm) (PDA; Accela™, Thermo Fisher Scientific Mississauga, ON, Canada), an evaporative light scattering detector (ELSD; Sedex, Sedere, Alfortville, France) and a high resolution mass spectrometer utilizing electrospray ionization (HRESIMS) (Orbitrap Exactive; Thermo Fisher Scientific, Mississauga, ON, Canada) (positive mode, monitoring m/z 200-2000). Chromatographic separation was achieved with a Kinetex 1.7 μm C.sub.18 100 Å 50×2.1 mm column (Phenomenex, Torrance, CA, USA) and a linear gradient from 95% H.sub.2O/0.1% formic acid (FA) (solvent A) and 5% acetonitrile (CH.sub.3CN)/0.1% FA (solvent B) to 100% solvent B over 5 min followed by a hold of 100% solvent B for 3 min with a flow rate of 400 μL/min. Examination of the ELSD chromatogram of the *V. paradoxus* RKNM-096 extract revealed five prominent peaks. The first peak eluted at 0.50 min and was present in the media blank indicating this peak was composed of media components. The following four peaks (1-4) eluted at 3.0 min, 5.04 min, 5.29 min and 5.39 min in the ELSD chromatogram, respectively. These peaks were not observed in the media blank extracts, indicating that these peaks were metabolic products of *V. paradoxus* RKNM-096. Peak 1 eluted at 3.00 min and examination of the mass spectrum of the corresponding peak in the total ion chromatogram (3.04 min) revealed the presence of two ions with mass to charge ratios (m/z) of 375.2855 and 397.2673, which is consistent with the anticipated [M+H].sup.+ and [M+Na].sup.+ for a compound with a molecular formula of C.sub.19H.sub.38N.sub.2O.sub.5 and mass of 374.2781. The mass spectra of peaks 2-4 were examined in an identical manner and the [M+H].sup.+ ions were identified as m/z 1007.6628, 1049.6778, and 1049.6734, respectively. The difference in mass between the [M+H].sup.+ ions associated with peaks 3 and 4 was 4.2 ppm, suggesting that these two compounds likely had an identical molecular formula, however the slight difference in retention time indicated that they were probably closely related structural analogues.

(90) The compounds were also elucidated using NMR. The NMR data indicated the presence of four carbonyl groups in addition to two sugar residues with characteristic anomeric carbon chemical shifts at δ.sub.C 101.4 and δ.sub.C 103.9. Key COSY and HMBC correlations allowed the chemical characterization of the amino acid-derived leucinol, a serine residue, and three 3-hydroxydecanoic acids (FIG. 1). The connectivity between the different moieties was further confirmed by tandem mass spectrometry. The two deoxyhexose residues were identified by

interpretation of ^1H - ^1H COSY correlations and coupling constant analysis. The small J-coupling exhibited by the anomeric proton H-1' OH 4.79, d, $J=1.4$ Hz) and the methine proton H-2' ($\delta_{\text{sub.H}} 3.86$, dd, $J=3.2, 1.4$ Hz) placed protons H-1' and H-2' in the equatorial position, while the larger J-coupling for H-4' ($\delta_{\text{sub.H}} 3.53$ - 3.48 , app. t) indicated the axial relationship with H-3' and H-5', and therefore suggested an α -rhamnopyranosyl residue. The HMBC cross peak between the anomeric proton H-1' and C-3C ($\delta_{\text{sub.C}} 76.5$) demonstrated the attachment of this sugar to the 3-hydroxydecanoic acid moiety. The second sugar residue was also identified as an α -rhamnopyranose on the basis of coupling constant values. The small J-coupling for H-1'' ($\delta_{\text{sub.H}} 5.01$, d, $J=1.5$ Hz) and H-2'' ($\delta_{\text{sub.H}} 3.98$, dd, $J=3.3, 1.5$ Hz) indicated the equatorial orientation of these protons, while the larger coupling constant for H-4'' ($\delta_{\text{sub.H}} 3.40$, app. t, $J=9.5$ Hz) demonstrated its axial relationship with H-3'' and H-5''. A key HMBC correlation between H-3' and C-1'' established a 1,3- α -glycosidic linkage between the two rhamnopyranose moieties.

(91) The structure of NB-RLP1006 is:

(92) ##STR00028##

(93) Organic extracts from *V. paradoxus* RKNM-096 were also fractionated by automated normal-phase chromatography followed by reversed-phase HPLC, which provided NB-RLP1048A and NB-RLP1048B. On the basis of HRESIMS analysis (NB-RLP1048A: HRESIMS m/z 1049.6778 $[\text{M}+\text{H}]^{\text{sup.}}$; NB-RLP1048B: HRESIMS 1049.6734 $[\text{M}+\text{H}]^{\text{sup.}}$, calcd for $\text{C}_{53}\text{H}_{97}\text{N}_{2}\text{O}_{18}$, 1049.6731), these compounds were determined to be mono-acetylated analogues of NB-RLP1006. The apparent molecular formula of these compounds is $\text{C}_{53}\text{H}_{96}\text{N}_{2}\text{O}_{18}$. Based on NMR analysis, NB-RLP1048A consisted of an inseparable mixture of acetylated glycolipopeptides with the structure:

(94) ##STR00029##

where any single R-group is an acetyl group, while all other R-groups are hydrogen atoms.

(95) The chemical structure of NB-RLP1048B was determined by 1D and 2D NMR spectroscopic techniques, confirming the location of the acetyl group at the C-3'' position.

(96) The chemical structure of NB-RLP1048B is:

(97) ##STR00030##

(98) The described fractionation scheme also yielded several other glycolipopeptide analogues produced by *V. paradoxus* RKNM-096 in smaller quantities, including 10.9 mg of NB-RLP978. The ^1H and ^{13}C NMR data were nearly identical to that of NB-RLP1006. The apparent molecular formula of NB-RLP978 is $\text{C}_{49}\text{H}_{90}\text{N}_{2}\text{O}_{17}$ (HRESIMS m/z 979.6307 $[\text{M}+\text{H}]^{\text{sup.}}$, calcd for $\text{C}_{49}\text{H}_{91}\text{N}_{2}\text{O}_{17}$, 979.6312). On the basis of tandem mass spectrometry, NB-RLP978 was determined to be an inseparable mixture of three closely related analogues, NB-RLP978A, NB-RLP978B, and NB-RLP978C, containing a C_8 acyl chain at one of the 3-hydroxyalkanoic acid positions.

(99) The chemical structure of NB-RLP978A-C is:

(100) ##STR00031##

where any single acyl chain is C_8 (i.e. $n=1, 2$, or $n=3$) while the remaining acyl chains are C_{10} (i.e. $n=5$). NB-RLP978A: $n=5, 5, 3$; NB-RLP978B: $n=5, 3, 5$; NB-RLP978C: $n=3, 5, 3$.

(101) The reversed-phase HPLC purification of NB-RLP1006 and NB-RLP978A-C also yielded NB-RLP950, an inseparable mixture of compounds with an apparent molecular formula of $\text{C}_{47}\text{H}_{86}\text{N}_{2}\text{O}_{18}$ (HRESIMS m/z 951.5982 $[\text{M}+\text{H}]^{\text{sup.}}$, calcd for $\text{C}_{47}\text{H}_{87}\text{N}_{2}\text{O}_{18}$, 951.5999), which is consistent with an analogue of NB-RLP1006 lacking four methylene groups. The ^1H NMR spectrum of NB-RLP950 was nearly identical to that of NB-RLP1006 and NB-RLP978. A ^{13}C spectrum was not obtained due to insufficient material. On the basis of tandem mass spectrometry, NB-RLP950 was determined to be a mixture of six closely related analogues: NB-RLP950A, NB-RLP950B, NB-RLP950C, NB-RLP950D, NB-RLP950E, and NB-RLP950F. These glycolipopeptide analogues either contain two

C.sub.8 acyl chain or one C.sub.6 acyl chain at the 3-hydroxyalkanoic acid positions.

(102) The chemical structure of NB-RLP950A-F is:

(103) ##STR00032##

where any two acyl chains are C.sub.8 (e.g. n.sub.1=n.sub.2=3 and n.sub.3=5) while the remaining acyl chain is C.sub.6 (i.e. n=1). NB-RLP950A: n.sub.1=5, n.sub.2=n.sub.3=3; NB-RLP950B: n.sub.2=5, n.sub.1=n.sub.3=3; NB-RLP950C: n.sub.3=5, n.sub.1=n.sub.2=3; NB-RLP950D: n.sub.1=n.sub.2=5, n.sub.3=1; NB-RLP950E: n.sub.1=n.sub.3=5, n.sub.2=1; NB-RLP950F: n.sub.2=n.sub.3=5, n.sub.2=1.

(104) The reversed-phase HPLC purification of NB-RLP1048B also yielded NB-RLP1020. The .sup.1H and .sup.13C NMR data of NB-RLP1020 were nearly identical to that of NB-RLP1048B. The apparent molecular formula of NB-RLP1020 is C.sub.51H.sub.92N.sub.2O.sub.18 (HRESIMS m/z 1021.6415 [M+H].sup.+ , calcd for C.sub.51H.sub.93N.sub.2O.sub.18, 1021.6418). On the basis of tandem mass spectrometry, NB-RLP1020 was determined to be an inseparable mixture of three closely related analogues, NB-RLP1020A, NB-RLP1020B, and NB-RLP1020C, comprising a C.sub.8 acyl chain at one of the 3-hydroxyalkanoic acid positions.

(105) The chemical structure of NB-RLP1020A-C is:

(106) ##STR00033##

where any single acyl chain is C.sub.8 (i.e. n.sub.1, n.sub.2, or n.sub.3=3) while the remaining acyl chains are C.sub.10 (i.e. n=5). NB-RLP1020A: n.sub.1=n.sub.2=5, n.sub.3=3; NB-RLP1020B: n.sub.1=n.sub.3=5, n.sub.2=3; NB-RLP1020C: n.sub.1=3, n.sub.2=n.sub.3=5.

(107) The reversed-phase HPLC fractionation also yielded an inseparable mixture of compounds with an apparent molecular formula of C.sub.51H.sub.92N.sub.2O.sub.18 (HRESIMS m/z 1021.6477 [M+H].sup.+ , calcd for C.sub.51H.sub.93N.sub.2O.sub.18, 1021.6418). Similar to NB-RLP1020A-C, the structure of these compounds is:

(108) ##STR00034##

where any single R-group is an acetyl group, while all other R-groups are hydrogen atoms and where any single acyl chain is C.sub.8 (i.e. n.sub.1, n.sub.2, or n.sub.3=3) while the remaining acyl chains are C.sub.10 (i.e. n=5).

(109) The reversed-phase HPLC fractionation also yielded NB-RLP1076. The .sup.1H and .sup.13C NMR data were nearly identical to that of NB-RLP1020A-C and NB-RLP1048B. The apparent molecular formula of NB-RLP1076 is C.sub.55H.sub.100N.sub.2O.sub.18 (HRESIMS m/z 1077.7046 [M+H].sup.+ , calcd for C.sub.55H.sub.101N.sub.2O.sub.18, 1077.7044). On the basis of tandem mass spectrometry, NB-RLP1076 was determined to be an inseparable mixture of three closely related analogues, NB-RLP1076A, NB-RLP1076B, and NB-RLP1076C, comprising a C.sub.12 acyl chain at one of the 3-hydroxyalkanoic acid positions.

(110) The chemical structure of NB-RLP1076A-C is:

(111) ##STR00035##

where any single acyl chain is C.sub.12 (i.e. n.sub.1, n.sub.2, or n.sub.3=7) while the remaining acyl chains are C.sub.10 (i.e. n=5). NB-RLP1076A: n.sub.1=n.sub.2=5, n.sub.3=7; NB-RLP1076B: n.sub.1=n.sub.3=5, n.sub.2=7; NB-RLP1076C: n.sub.1=7, n.sub.2=n.sub.3=5.

(112) The reversed-phase HPLC fractionation also yielded an inseparable mixture of compounds with an apparent molecular formula of C.sub.55H.sub.100N.sub.2O.sub.18 (HRESIMS m/z 1077.7098 [M+H].sup.+ , calcd for C.sub.55H.sub.101N.sub.2O.sub.18, 1077.7044). Similar to NB-RLP1076A-C, the structure of these compounds is:

(113) ##STR00036##

where any single R-group is an acetyl group, while all other R-groups are hydrogen atoms and where any single acyl chain is C.sub.12 (i.e. n.sub.1, n.sub.2, or n.sub.3=7) while the remaining acyl chains are C.sub.10 (i.e. n=5).

(114) Using portions of the *V. paradoxus* RKNM-096 glycolipopeptide biosurfactant biosynthetic gene cluster as in silico probes against published bacteria genomes (described below), we identified

Janthinobacterium agaricidamnorum DSM 9628 as a potential producer of glycolipopeptide biosurfactants similar to those isolated from *V. paradoxus* RKNM-096. *J. agaricidamnorum* was cultured and extracted as described above for *V. paradoxus* RKNM-096 and the resulting organic extract (110.4 mg) of was subjected to automated reversed-phase chromatography with a RediSep C.sub.18 column using a H.sub.2O/CH.sub.3OH gradient. Fractions containing the glycolipopeptide (77.6 mg) were combined and a portion of this material was subjected to further separation by reversed-phase HPLC, which yielded 17.1 mg of NB-RLP860 and 6.4 mg of NB-RLP832. Analysis of NB-RLP860 by HRESIMS (HRESIMS m/z 861.6033 [M+H].sup.+, calcd for C.sub.45H.sub.85N.sub.2O.sub.13, 861.6046) indicated an apparent molecular formula of C.sub.45H.sub.84N.sub.2O.sub.13 and five degrees of unsaturation. The .sup.1H and .sup.13C NMR data of NB-RLP860 were similar to NB-RLP1006, except the NMR spectra lacked resonances belonging to the second α -rhamnopyranose moiety.

(115) The chemical structure of NB-RLP860 was determined by 1D and 2D NMR spectroscopy. The structure of NB-RLP860 is:

(116) ##STR00037##

(117) Analysis of NB-RLP832 by HRESIMS (HRESIMS m/z 833.5734 [M+H].sup.+, calcd for C.sub.45H.sub.81N.sub.2O.sub.13, 833.5733) indicated an apparent molecular formula of C.sub.43H.sub.80N.sub.2O.sub.13. The .sup.1H and .sup.13C NMR data were nearly identical to that of NB-RLP860. On the basis of tandem mass spectrometry, NB-RLP832 was determined to be an inseparable mixture of three closely related analogues, NB-RLP832A, NB-RLP832B, and NB-RLP832C, comprising a C.sub.8 acyl chain at one of the 3-hydroxyalkanoic acid positions.

(118) The chemical structure of NB-RLP832A-C is:

(119) ##STR00038##

where any single acyl chain is C.sub.8 (i.e. n.sub.1, n.sub.2, or n.sub.3=3) while the remaining acyl chains are C.sub.10 (i.e. n=5). NB-RLP832A: n.sub.1=n.sub.2=5, n.sub.3=3; NB-RLP832B: n.sub.1=n.sub.3=5, n.sub.2=3; NB-RLP832C: n.sub.1=3, n.sub.2=n.sub.3=5.

(120) Glycolipopeptides NB-RLP860 and NB-RLP832A-C were also detected in small quantities in organic extracts of *V. paradoxus* RKNM-096 by LC-MS analysis. Analysis of HRESIMS chromatograms revealed [M+H].sup.+ ions of m/z 861.6073 and m/z 833.5749, which are consistent with the predicted m/z of [M+H].sup.+ ions for NB-RLP860 (calcd for C.sub.45H.sub.85N.sub.2O.sub.13, m/z 861.6046 [M+H].sup.+) and NB-RLP832A-C (calcd for C.sub.45H.sub.81N.sub.2O.sub.13, m/z 833.5733 [M+H].sup.+).

(121) Analysis of organic extracts of *V. paradoxus* RKNM-096 also revealed three peaks in the HRESIMS chromatogram exhibiting [M+H].sup.+ ions of m/z 903.6213, which is consistent with the predicted [M+H].sup.+ ions for an acetylated analogue of NB-RLP860 (m/z 903.6152 [M+H].sup.+). As these compounds were produced in small quantities, attempts to determine their structures unambiguously by NMR spectroscopy were prohibited. These compounds were not detected in organic extracts from *J. agaricidamnorum* DSM 9628. Given the observed fragment ions of m/z 715.5480 (b) and 598.4310 (bf), these compounds were identified as acetylated glycolipopeptides NB-RLP902 with the structure:

(122) ##STR00039##

where any single R-group is an acetyl group, while all other R-groups are hydrogen atoms.

(123) Fractions generated by automated reversed-phase chromatography of organic extracts from *V. paradoxus* RKNM-096 were enriched with NB-RLP902. Also detected in the HRESIMS chromatograms of these fractions was a small peak exhibiting a [M+H].sup.+ ion of m/z 875.5888, which is consistent with an analogue of NB-RLP902 lacking two methylene groups. This [M+H].sup.+ ion was not observed in organic extracts from *J. agaricidamnorum* DSM 9628. The observed fragment ion of 687.5164 (b) indicates that this compound is also a glycolipopeptide. Similar to NB-RLP978A-C, NB-RLP1020A-C, and NB-RLP832A-C, it is proposed that this peak is comprised of three compounds NB-RLP874A-C with the structure:

(124) ##STR00040##

where any single R-group is an acetyl group, while all other R-groups are hydrogen atoms and where any single acyl chain is C.sub.8 (i.e. n.sub.1, n.sub.2, or n.sub.3=3) while the remaining acyl chains are C.sub.10 (i.e. n=5).

Example 4: Deacetylation of NB-RLP1048A and Other Acetylated Glycolipopeptide Biosurfactants Produced by *Variovorax paradoxus* RKNM-096

(125) It is known that the relative amount of NB-RLP1006 and acetylated glycolipopeptides (e.g. NB-RLP1048A) produced by *V. paradoxus* RKNM-096 may vary between batches using different culture media and fermentation conditions. As a result, the surfactant properties of the extracted glycolipopeptide product may also vary. As product consistency is important to be competitive in the biosurfactant industry, a method to selectively remove the acetate from R.sub.5a, R.sub.5b, R.sub.6a, R.sub.6b, R.sub.7a, R.sub.7b, R.sub.10, and R.sub.11 was developed to generate a consistent glycolipopeptide product comprised of NB-RLP1006 with >95% purity by weight (Scheme 1). The method utilizes NaOH within a narrow concentration range to selectively remove acetate moieties without inducing further hydrolysis of the amide, ester, or glycosidic linkages of the glycolipopeptide. The NaOH concentration and reaction solvent both have a demonstrated role in controlling the extent of hydrolysis and achieving selectivity. Optimal NaOH concentrations are directly proportional to the concentration and composition of the glycolipopeptides in the reaction medium. Reaction solvents with higher water composition, such as H.sub.2O:acetone (9:1), showed better selectivity and minimized the hydrolysis of the ester linkages between the β -hydroxyalkanoic acid moieties.

(126) ##STR00041##

(127) It is known that deacetylation of the glycolipopeptide mixture may be achieved with variations to the method described herein. It is possible that inorganic bases other than NaOH, including but not limited to LiOH, KOH, Na.sub.2CO.sub.3, NH.sub.3, and NH.sub.4OH, or organic bases, including but not limited to tetrabutylammonium hydroxide or alkylamines, may be utilized. The selective deacetylation may also be achieved enzymatically using esterases, including but not limited to acylesterases and lipases.

(128) Hydrolysis of the glycolipopeptide mixture is known to produce several products, including but not limited to the lipopeptides NB-RLP356 (HRESIMS m/z 357.2745 [M+H].sup.+, calcd for C.sub.19H.sub.37N.sub.2O.sub.4, 357.2748; m/z 379.2567 [M+Na].sup.+, calcd for C.sub.19H.sub.36N.sub.2O.sub.4Na, 379.2565), NB-RLP374 (HRESIMS m/z 375.2851 [M+H].sup.+, calcd for C.sub.19H.sub.39N.sub.2O.sub.5, 375.2854), and NB-RLP526 (HRESIMS m/z 527.4054 [M+H].sup.+, calcd for C.sub.29H.sub.55N.sub.2O.sub.6, 527.4055), and the glycolipids NB-RLP480 (HRESIMS m/z 481.2599 [M+H].sup.+, calcd for C.sub.22H.sub.41O.sub.11, 481.2643; m/z 503.2465 [M+Na].sup.+, calcd for C.sub.22H.sub.40O.sub.11Na, 503.2463) and NB-RLP650 (HRESIMS m/z 651.3962 [M+H].sup.+, calcd for C.sub.32H.sub.59O.sub.13, 651.3950). Given their amphiphilic structures, these compounds are also expected to behave as surface active agents and may exhibit surfactant properties that may be unique or complementary to the glycolipopeptides. These compounds are known to be formed during the deacetylation process described herein and are thus present in the glycolipopeptide final product. Although normally present in small quantities (<5% by weight), these compounds may contribute to the surfactant characteristics of the glycolipopeptide product. Hydrolysis of the glycolipopeptides may also occur spontaneously, for instance during the extraction and purification, to generate these compounds. For instance, the lipopeptide NB-RLP374 is detected in the organic extract of *V. paradoxus* RKNM-096 before the glycolipopeptide material is subjected to any downstream modification.

(129) The chemical structure of NB-RLP356 is:

(130) ##STR00042##

(131) The chemical structure of NB-RLP374 is:

(132) ##STR00043##

(133) The chemical structure of NB-RLP526 is:

(134) ##STR00044##

(135) The chemical structure of NB-RLP480 is:

(136) ##STR00045##

(137) The chemical structure of NB-RLP650 is:

(138) ##STR00046##

Example 5: Surface Activity

(139) As summarized in Table 1, the critical micelle concentrations (CMCs) of NB-RLP1006, NB-RLP978, NB-RLP860, and NB-RLP-1048B were determined by the Du Nouy method utilizing a Kibron Delta-8 multichannel microtensiometer (Kibron Inc., Helsinki, Finland). All samples were prepared in degassed deionized water (Millipore, Etobicoke, ON, CA) at concentrations ranging from 0 to 2.0 mM. All measurements were recorded between 24 and 25° C. and performed in duplicate. The critical micelle concentration of both NB-RLP1006 and NB-RLP978 was 0.20 mM (0.02 wt %). Surface tension measurements indicated that NB-RLP1006 and NB-RLP978 were capable of reducing the surface tension of water from 72 to 35.5 mN/m at their CMC. Meanwhile, NB-RLP860 and NB-RLP1048B exhibited CMC values of 0.85 mM (0.07 and 0.09 wt %, respectively), reducing the surface tension of water to 36.2 and 36.9 mN/m, respectively. The surface activity of NB-RLP1006 was compared to rhamnolipids A and B, which were purified from a commercially available rhamnolipid mixture (R90; AGAE Technologies, Corvallis, OR, USA) by reversed-phase HPLC. Rhamnolipids A and B both exhibited a CMC of 0.06 mM (0.003 and 0.004 wt %, respectively) in which the surface tension of water was reduced to 28.2 and 39.0 mN/m, respectively. The higher CMC values for NB-RLP860 and NB-RLP1048B may be due to their poor aqueous solubility.

(140) TABLE-US-00001 TABLE 1 Surfactant properties of isolated glycolipopeptides compared to rhamnolipids. Critical micelle concentration (CMC) and surface tension reduction of water are shown. Minimum Surface CMC Tension Compound (mM) (mN/m) NB-RLP1006 0.20 35.5 NB-RLP1048B 0.85 36.9 NB-RLP860 0.85 36.2 NB-RLP978 0.20 35.5 Rhamnolipid A 0.06 28.2 Rhamnolipid B 0.06 39.0

(141) The characteristic curvature (C_c) of NB-RLP1006 was determined using the hydrophilic-lipophilic difference-net average curvature (HLD-NAC) model to calculate the shift in chemical potential when NB-RLP1006 is transferred from the oil to the aqueous phase as a function of salinity by the following general equation:

$$HLD = F(S) - k \times EACN + F(A) - \alpha \times \Delta T + C_c$$

where $F(S)$ is a function of salinity, k is a coefficient equal to 0.17, EACN (effective alkane carbon number) is the number of carbons in the alkane oil phase, α is a coefficient dependent on the type of surfactant (ionic, ethoxylates, etc), and ΔT is the effect of temperature. Four mixtures of NB-RLP1006 and sodium dihexyl sulfosuccinate (SDHS) were prepared with a total surfactant concentration of 1.8 mg/mL using the following NB-RLP1006/SDHS ratios: 0, 12, 24, and 40 wt % NB-RLP1006. An electrolyte scan was performed for each mixture by varying the NaCl concentration from 0 to 6.0% (w/v). Each mixture was added to an equal volume of toluene, which constituted the oil phase, and shaken vigorously. The optimal salinity (S^*) was identified as the concentration of NaCl in which a Winsor Type III microemulsion was formed, wherein the separate middle phase was composed of an equal volume of oil and water. A plot of the NB-RLP1006/SDHS molar ratios versus S^* was generated and C_c was calculated from the line of best fit. The C_c value for NB-RLP1006 was determined to be +5.2, a value that reflects the hydrophobic nature of this biosurfactant.

(142) The emulsifying properties of NB-RLP1006 were determined using the emulsification index as described above. Pure NB-RLP1006 exhibited strong emulsification activity with an E.sub.24 value of 53% at 1 mg/mL in deionized water. The emulsification of NB-RLP1006 is pH-dependent

with E.sub.24 values of 8, 38, and 31% at pH 3, 6, and 8, respectively. The type of emulsion formed by NB-RLP1006 (e.g. oil-in-water or water-in-oil) was determined using the drop dilution test. An emulsion was formed by vigorously mixing a 1 mg/mL solution of RLP1006 in deionized water with an equal volume of kerosene for 1 min. A portion (20 μ L) of the emulsion was transferred to 0.5 mL of deionized water and 0.5 mL of kerosene and dilution of the emulsion in each liquid was monitored. The emulsion formed by NB-RLP1006 was readily dispersed in the aqueous phase, indicating that the continuous phase of the emulsion was water and that an oil-in-water (o/w) emulsion was formed by NB-RLP1006 under these conditions.

(143) These results established that NB-RLP1006 is a potent biosurfactant capable of lowering the surface tension of water to 35.5 mN/m with a CMC comparable to that of two other well-characterized biosurfactants, rhamnolipids A and B. NB-RLP1006 also exhibits strong emulsification activity forming o/w emulsions under the conditions described herein.

Example 6: Cytotoxicity Testing of the Glycolipopeptides

(144) To evaluate the safety profile of the glycolipopeptides, cytotoxicity testing was conducted against two normal human cell lines, BJ fibroblast cells ATCC CRL-2522 and adult epidermal keratinocytes (HEKa; Life Technologies, Carlsbad, CA, USA). BJ fibroblasts were grown and maintained in 15 mL Eagle's minimal essential medium supplemented with fetal bovine serum (10% v/v), penicillin (100 μ U) and streptomycin (100 μ g/mL). HEKa cells were grown and maintained in 15 mL of EPI life medium (Life Technologies, Carlsbad, CA, USA) supplemented with HKGS growth supplement (10% v/v; Life Technologies, Carlsbad, CA, USA) and 50 μ g/mL gentamicin (Sigma-Aldrich, St. Louis, MO, USA). Cells were cultured in T75cm.sup.2 cell culture flasks and incubated at 37° C. in a humidified atmosphere of 5% CO.sub.2. For BJ fibroblasts culture media was refreshed every two to three days and cells were not allowed to exceed 80% confluence. For HEKa cells growth medium was refreshed every 2 d until the cells reached 50% confluence and then the medium was refreshed every 24 h until 80% confluence was obtained. At 80% confluence, the cells were counted, diluted to 10,000 cells/well in growth medium lacking antibiotics and 90 μ L of cell suspension was transferred into the wells of 96-well treated cell culture plates. The plates were incubated as before to allow cells to adhere to the plates for 24 h before treatment. DMSO was used as the vehicle at a final concentration of 1%. All compounds tested were re-solubilized in DMSO and a dilution series was prepared for each cell line using the respective cell culture growth medium, 10 μ L of which were added to the assay wells yielding eight final concentrations ranging from 512 μ g/mL to 8 μ g/mL per well (final well volume of 100 μ L). The fibroblasts and HEKa cells were incubated as previously described for 24 h. All samples were tested in triplicate. Each plate contained four un-inoculated media blanks (media+1% DMSO), four untreated growth controls (media+1% DMSO+cells), and one column containing a serially diluted zinc pyrithione positive control. AlamarBlue (Life Technologies, Carlsbad, CA, USA) was added to each well 24 h after treatment (10% v/v). Fluorescence (560/12 excitation, 590 nm emission) was monitored using a Varioskan Flash Multimode plate reader both at time zero and 4 h after the addition of alamarBlue. After subtraction of fluorescence at time zero from 4 h readings the percentage of cell viability relative to vehicle control wells was calculated. Low cytotoxic activity was displayed against the HEKa and BJ fibroblast cell lines. The observed IC.sub.50 and MIC.sub.90 values for the glycolipopeptides were significantly higher than the positive control zinc pyrithione, which served as an industry benchmark for topical antimicrobial agents (Table 2). These results indicate that the glycolipopeptides exhibit low cytotoxicity towards human skin cells and thus may be safe for use in applications which result in dermal contact such as cosmetic products.

(145) TABLE-US-00002 TABLE 2 Cytotoxicity testing results for the glycolipopeptides. Values indicate the half maximal inhibitory concentrations (IC.sub.50) and minimum inhibitory concentration that results in 90% of growth inhibition (MIC.sub.90) in μ g/mL. Error is reported as standard deviation. Eukaryotic Cells HEKa HEKa BJ BJ Compound (IC.sub.50) (MIC.sub.90)

(IC.sub.50) (MIC.sub.90) NB-RLP1006 15.5 ± 1.7 64-128 19.5 ± 2.4 32 NB-RLP1048B 19.3 ± 4.0
 64-128 18.7 ± 1.6 32 NB-RLP860 15.5 ± 1.6 128 16.3 ± 0.3 32 Zinc pyrithione 0.20 ± 0.001 1
 2.2 ± 0.3 4

Example 7: Sequencing of the *V. paradoxus* RKNM-096 Glycolipopeptide and Rhamnose Biosynthetic Gene Clusters

(146) To establish the genetic basis for the biosynthesis of the novel glycolipopeptide biosurfactants described here, the genome of *V. paradoxus* RKNM-096 was sequenced. *V. paradoxus* RKNM-096 was cultured in ISP2 broth and genomic DNA was isolated using the UltraClean® Microbial DNA Isolation Kit according to the manufacturer's recommendations (Mo Bio, Carlsbad, CA, USA). The genome was sequenced at the McGill University and Genome Quebec Innovation Centre (Montreal, QC, CA) using 2 SMRT Cells in a PacBio RSII sequencer (Pacific Biosciences, Menlo Park, CA, USA). A total of 140, 476 raw subreads with an average length of 11,269 bp were generated and genome assembly was achieved using a HGAP workflow (Chin et al. [2013] *Nature Methods* 10, 563). Briefly, raw subreads were generated from raw .bas.h5 PacBio data files. A subread length cutoff value ($30\times$) was extracted from subreads and used in the preassembly (BLASR) step, which consists of aligning short subreads on long subreads (Chaisson and Tesler [2012] *BMC Bioinformatics* 13, 238). Since errors in PacBio reads are random, the alignment of multiple short reads on longer reads enables correction of sequencing errors on long reads. These long corrected reads were then used as seeds in a subsequent assembly prepared using the Celera assembler (Myers et al. [2000] *Science* 287, 2196), which generates contigs. These contigs were then 'polished' by aligning raw reads on contigs (BLASR) which were then processed through a variant calling algorithm (Quiver) that generates high quality consensus sequences using local realignments and PacBio quality scores (Chin et al. [2013] *Nature Methods* 10, 563). Over 161,717,463 bp of corrected long subreads were obtained and resulted in the assembly of two contigs. One contig contained 7,193,071 bp while the other contained 1,767 bp. The genome was annotated using the RAST server (Aziz et al. [2008] *BMC Genomics* 9, 75; Overbeek et al. [2014] *Nucleic Acid Res.* 42, D206; Brettin et al. [2015] *Sci Rep.* 5, 8265). The function of open reading frames (ORFs) identified by the RAST annotation were further explored by BLASTP (Altschul et al. [1997] *Nucleic Acids Res.* 25, 3389) and conserved domain (Marchler-Bauer and Bryant [2004] *Nucleic Acids Res.* 32, W327) analysis of deduced amino acid sequences. (147) Based on the structure of NB-RLP1006 it was hypothesized that its biosynthesis would require a NRPS to synthesize the dipeptide, one or more acyltransferases to acylate the peptide and generate the 3-(3-(3-hydroxydecanoyloxy) decanoyloxy) decanoyl moiety and one or more glycosyltransferases. Scanning the genome for genes encoding NRPSs identified two loci. One locus contained a single NRPS-encoding gene followed by two glycosyltransferases, thus this locus (12,721 bp) was analyzed further. Six ORFs were identified in this locus, which were predicted to play an integral role in glycolipopeptide biosynthesis (Table 3). The six genes, designated rlpA to rlpF, are oriented in the same direction and form a contiguous region in the *V. paradoxus* RKNM-096 genome.

(148) TABLE-US-00003 TABLE 3 Deduced functions of Orfs identified in the *V. paradoxus* RKNM-096 glycolipopeptide (Seq. ID: 1) and dTDP-L-rhamnose biosynthetic gene clusters (Seq. ID: 2). Seq. ID. Seq. ID. (DNA) Source Name Start Stop (protien) Size (aa) Proposed Function
 3 Seq. ID: 1 rlpA 121 1035 4 304 LysR transcriptional regulator 5 Seq. ID: 1 rlpB 1437 8912 6 2491 Nonribosomal peptide synthetase 7 Seq. ID: 1 rlpC 8924 10243 8 439 dTDP-rhamnosyl transferase 9 Seq. ID: 1 rlpD 10276 10488 10 70 MbtH protein 11 Seq. ID: 1 rlpE 10497 11465 12 322 dTDP-rhamnosyl transferase 13 Seq. ID: 1 rlpF 11462 12721 14 419 MFS transporter 15 Seq ID: 2 rmlB 299 1378 16 359 dTDP-glucose 4,6-dehydratase 17 Seq ID: 2 rmlD 1375 2265 18 296 dTDP-4-dehydrorhamnose reductase 19 Seq ID: 2 rmlA 2298 3194 20 298 Glucose-1-phosphate thymidyltransferase 21 Seq ID: 2 rmlC 3191 3736 22 181 dTDP-4-dehydrorhamnose 3,5-epimerase

(149) Genes involved in regulation. The first gene, *rlpA*, encodes a protein that exhibits similarity to transcriptional regulators belonging to the LysR family. Conserved domain analysis indicated that RlpA contained an amino-terminal helix-turn-helix domain and a carboxy-terminal LysR substrate binding domain, which is consistent with the domain architecture of LysR transcriptional regulators. This family of regulators can function as transcriptional activators or repressors (Maddocks and Oyston [2009] *Microbiology* 154, 3609), thus it is likely that RlpA plays a role in the regulation of glycolipopeptide biosynthesis.

(150) Genes involved in peptide biosynthesis. Following *rlpA* is large gene, *rlpB*, (7,476 bp), which encodes a NRPS. Domain analysis (Bachmann and Ravel [2009] *Meth. Enzymol.* 458, 181) indicated that the NRPS consists of two modules (M1 and M2) with the following domain organization (C-A-PCP).sub.M1-(C-A-PCP-R).sub.M2. The dimodular structure and domain organization suggests that RlpB generates a dipeptide, which is consistent with structure of the *V. paradoxus* RKNM-096 glycolipopeptides. The first domain of the first module of RlpB is a condensation domain. The presence of a C-domain at the beginning of a NRPS initiation module is characteristic of acylated peptides. Amino-terminal C-domains can catalyze amide bond formation between the first amino acid of a peptide and a fatty acid. The fatty acid can be presented to the C-domain as an acyl-ACP intermediate, as in the case of CDA biosynthesis (Kopp et al. [2008] *J. Am. Chem. Soc.* 130, 2656), or an acyl-CoA intermediate, as in the case of surfactin biosynthesis (Krass et al. [2010] *Chem. Biol.* 17, 872). A phylogenetic analysis of the RlpB initiation module C-domain (residues 12-437) was conducted using the NaPDoS program (Ziemert et al. [2012] *PLoS One* 7, e34064). The RlpB domain clustered closely with C-domains from initiation modules that catalyze the condensation of a fatty acid precursor with an amino acid. The most closely related C-domain in the NaPDoS reference database was the initiation module of the bacillibactin NRPS (38% identity), which catalyzes the condensation of 2,3-dihydroxybenzoyl-ACP with glycine (May et al. [2001] *J. Biol. Chem.* 278, 7209). This suggests that glycolipopeptide biosynthesis starts with the condensation of a fatty acid with the first amino acid of the peptide (serine). Similar analysis of the second C-domain indicated it was most closely related to the second C-domain of the bacillibactin dimodular NRPS, Dhbf (54% identity). Phylogenetic analysis revealed that the M2 C-domain of RlpB clustered with C-domains catalyzing the condensation of two L-amino acids (Ziemert et al. [2012] *PLoS One* 7, e34064), which is consistent with the glycolipopeptide structure.

(151) To predict the substrate specificity of the RlpB A-domains, the substrate specificity codes were extracted from the A-domain active sites (8 residues between motifs A3 and A6) and compared to known A-domain specificity codes using the NRPS Predictive Blast tool (Bachmann and Ravel [2009] *Meth. Enzymol.* 458, 181). The specificity code of the M1 A-domain was most similar to A-domains from the nostopeptolide, pyoverdine, CDA and enterobactin NRPSs that activate L-serine (75-87% identity, 87-100% similarity, E-value 0.023-0.039), suggesting that L-serine is incorporated by M1. This observation is consistent with the structure of the glycolipopeptides. The M2 A-domain specificity code showed low homology (50% identity, 100% similarity, E-value 0.98) to an A-domain of the tyrocidine NRPS (TycB), which activates L-phenylalanine or L-tryptophan (Mootz and Marahiel [1997] *J. Bacteriol.* 179, 6843). This low level of similarity precludes prediction of the substrate specificity of this A-domain. Based on the structure of the *V. paradoxus* RKNM-096 glycolipopeptides the second A-domain would be expected to activate L-leucine. Comparison of the A-domain specificity code of RlpB module 2 to leucine specificity codes (Stachelhaus et al. [1999] *Chem. Biol.* 6, 493) also revealed low similarity, thus the RlpB M2 A-domain specificity code may represent a novel variant for leucine, although biochemical evidence would be needed to establish the substrate specificity of this domain. The PCP domains of RlpB were also analyzed and both were found to contain the core PCP domain motif with an invariant serine which represents the 4'-phosphopantetheine attachment site (Konz and Marahiel [1999] *Chem. Biol.* 6, R39).

(152) The final domain of RlpB is an R-domain. R-domains utilize NAD(P)H as a co-factor to

reductively release PCP-bound final products as an aldehyde or alcohol (Du and Lou [2010] *Nat. Prod. Rep.* 27, 255). The presence of a leucinol residue at the carboxy-terminus of the glycolipopeptide dipeptide moiety is consistent with release of an acylated dipeptide intermediate by an R-domain. Collectively, the domain structure and organization of RlpB, as well as the predicted substrate specificity of the individual domains are consistent with the structure of the glycolipopeptides produced by *V. paradoxus* RKNM-096.

(153) A small gene (rlpD) encoding a 70 amino acid protein that shows similarity to MbtH-like proteins was found downstream of rlpB. These proteins are often found in association with NRPSs and have been demonstrated to be essential for non-ribosomal peptide the production. (Baltz [2014] *J. Ind. Microbiol. Biotechnol.* 41, 357). Recently these proteins have been shown to facilitate adenylation reactions via direct interaction with A-domains (Herbst et al. [2013] *J. Biol. Chem.* 288, 1991). Thus we predict that RlpD interacts with one or both A-domains of RlpB to facilitate dipeptide formation.

(154) Genes involved in glycosylation. Glycosylation of the acylated dipeptide generated by RlpB is likely catalyzed by two ORFs (rlpC and E) downstream of rlpB. The deduced amino acid sequence of rlpC (439 aa) shows similarity to the GT1 family of glycosyltransferases, which utilize activated sugars as substrates to transfer sugar moieties to a diverse array of acceptor molecules (Breton et al. [2006] *Glycobiology* 16, 29R). The deduced amino acid sequence of rlpE (322 aa) shows similarity to dTDP-rhamnosyltransferases. In rhamnolipid biosynthesis two glycosyltransferases are utilized to sequentially transfer two rhamnosyl units to the lipid component of rhamnolipid (Deziel et al. [2003] *Microbiology* 149, 2005). RhlB transfers rhamnose from dTDP-L-rhamnose to the free β -hydroxyl group of 3-(3'-hydroxydecanoyloxy)decanoic acid (HDD) to generate mono-RL, while di-RL is formed by the transfer of an additional rhamnose from dTDP-L-rhamnose to mono-RL by RhlC (Abdel-Mawgoud et al. [2011] in *Biosurfactants*, Springer-Verlag, Berlin Heidelberg). The relationship between RlpC and RlpE and the RhlB and RhlC homologs from *P. aeruginosa* PAO1, *B. thialandensis* E264 and *B. pseudomallei* 1710B was investigated via the generation of a phylogenetic tree (unweighted pair group method with arithmetic mean method). In this analysis RlpC clustered with the RhlB orthologs while RlpE clustered with the RhlC orthologs. While RlpC clustered with the RhlB orthologs, it did not cluster tightly as it showed limited sequence identity with these enzymes (18.6-23.1%). In contrast, RlpE shared between 39.6-40.7% identity with the RhlC orthologs. This data suggests that RlpC and RlpE perform similar functions as RhlB and RhlC, respectively. We hypothesize that RlpC catalyzes the rhamnosylation of an acylated dipeptide intermediate utilizing dTDP-L-rhamnose as the carbohydrate donor. The limited sequence homology between RlpC and the RhlB orthologs may reflect the significant difference in glycosylation substrates utilized by the enzymes. RlpE is predicted to catalyze the second glycosylation reaction, transferring rhamnose from dTDP-L-rhamnose to the RlpC reaction product.

(155) Genes encoding dTDP-L-rhamnose biosynthesis were not found in close proximity to the glycolipopeptide gene cluster. Scanning the genome for homologs of *P. aeruginosa* PAO1 rhamnose biosynthetic genes (rmlBDAC) identified four genes that exhibited strong sequence similarity to those from *P. aeruginosa* (identity/similarity: RmlB—79%/89%, RmlD—60%/71%, RmlA—78%/89%, RmlC—66%/80%). In the *V. paradoxus* RKNM-096 genome the four genes are clustered and are found in the same order as in *P. aeruginosa* (rmlBDAC) (Rahim et al. [2000] *Microbiology* 146, 2803). This locus likely provides the dTDP-L-rhamnose substrates utilized by RlpC and RlpE. Modulation of expression of one or more components of the dTDP-L-rhamnose biosynthetic pathway by one skilled in the art may be an effective approach to increase glycolipopeptide yields.

(156) Genes involved in transport. Directly downstream of rlpE is an ORF (rlpF) encoding a protein, which is similar to major facilitator superfamily transporters from a variety of bacteria. RlpF exhibits 38% identity and 54% similarity to PA1131 from *P. aeruginosa* PAO1, which is

immediately upstream of rhlC (Dubeau [2009] *BMC Microbiol* 9, 263). RlpF is likely involved in glycolipopeptide efflux.

(157) Genes involved in the biosynthesis of the lipid moiety. In rhamnolipid biosynthesis the HDD moiety is produced by RhlA, which condenses two β -hydroxydecanoyl-ACP molecules from fatty acid biosynthesis to yield 3-(3'-hydroxydecanoyloxy)decanoic acid. Scanning of the *V. paradoxus* RKNM-096 genome for RhlA homologs did not identify any proteins with significant similarity to RhlA. Thus generation of the lipid moiety of the RKNM-096 glycolipopeptides is likely directed by a novel, yet to be identified mechanism.

(158) Genes involved in glycolipopeptide acetylation. Acetylated analogues of NB-RLP1006 are abundant in *V. paradoxus* RKNM-096 fermentation broths. No genes encoding acetyltransferases were identified in the gene cluster. Thus it is likely that acetylation is catalyzed by an enzyme encoded elsewhere in the *V. paradoxus* RKNM-096 genome.

(159) Proposed biosynthesis. Glycolipopeptide biosynthesis presumably starts with the formation of the 3-(3-(3-hydroxydecanoyloxy)decanoyloxy)decanoyl moiety via a yet to be identified mechanism. After formation of the lipid moiety it is likely presented to the C-domain of RlpB M1 which condenses the lipid moiety with L-serine. RlpB M2 then incorporates L-leucine to form a PCP-bound acylated dipeptide intermediate which is released from the enzyme by the C-terminal R-domain of RlpB, resulting in the formation of a terminal L-leucinol residue. dTDP-L-Rhamnose, produced by the rmlBDAC operon, is then utilized by the rhamnosyltransferases RlpC and RlpE to sequentially glycosylate the aglycone resulting in the production of the final glycosylated glycolipopeptide NB-RLP1006. NB-RLP1006 would serve as a substrate for acetylation to form NB-RLP1048A and NB-RLP1048B.

(160) To prove the involvement of the rlpA-rplF gene cluster in the biosynthesis of glycolipopeptides in *V. paradoxus* RKNM-096 rlpE was expressed in *E. coli* and the activity of the enzyme demonstrated using NB-RPL860 as a substrate. Bioinformatics analysis indicated RlpE catalyzes the second rhamnosylation in glycolipopeptide biosynthesis, converting mono-rhamnosylated glycolipopeptides (e.g. NB-RLP832 and NB-RLP860) to di-rhamnosylated glycolipopeptides (e.g. NB-RLP978 and NB-RLP1006). The rlpE gene was cloned in pET28a (EMD Millipore, Darmstadt, DE) with an amino-terminal hexa-histidine tag using standard cloning techniques and mutation-free cloning was verified by sequencing. Due to the high GC content of rlpE, *E. coli* Rossetta DE3 pLysS (EMD Millipore) was chosen as the expression host as this strain expresses tRNAs for rare GC-rich codons (AGG, CCA, GGA). A single colony was used to inoculate 50 mL of LB Miller (EMD Millipore) supplemented with 50 μ g/mL of kanamycin (Sigma-Aldrich) and 34 μ g/mL of chloramphenicol (Sigma-Aldrich) and the flask was incubated at 37° C. with shaking at 250 rpm overnight. Expression cultures (50 mL) were performed in LB Miller supplemented with kanamycin and chloramphenicol. These cultures were inoculated with 0.5 mL of the overnight culture and cultured at 37° C. and 250 rpm until the optical density (600 nm) reached 0.5, following which IPTG was added to a final concentration of 1.0 mM to induce protein expression and the cultures were incubated at 15° C. for 24 h. Cells were harvested by centrifugation (6 000 $\times$ g for 5 min) and washed once with 20 mM Tris-HCl (pH 8.0). The cell pellet was frozen at -80° C. until purification could be performed. To purify His-tagged RlpE, the cells were thawed, suspended in lysis buffer (500 mM NaCl, 5% glycerol, 1% Triton X-100, 25 mM Tris-HCl, pH 8.0) and then lysed via sonication. Cell debris and insoluble protein was removed by centrifugation at 15 000 $\times$ g for 30 min. The supernatant was mixed with 0.5 mL of HisPur Ni-NTA resin (Thermo Fisher Scientific). The resin was washed six times with 1.0 mL of 75 mM imidazole. His-tagged RlpE was eluted with 1.0 mL of 250 mM imidazole. Four batch elutions were performed and pooled. The imidazole elution buffer was exchanged with enzyme buffer (25 mM Tris-HCl, 10% glycerol) and concentrated by centrifugal filtration using a Macrosep 3 kDa spin filter (Pall). Following concentration the enzyme was aliquoted and stored at -80° C. The purity of the enzyme was analyzed by denaturing polyacrylamide gel electrophoresis (4-15% Mini-

PROTEAN precast gel, 160 V, 30 min; Bio-Rad). The calculated molecular weight of His-tagged RlpE was 38.2 KDa. The apparent molecular weight of the purified protein was 33.05 kDa, which was in good agreement with the expected molecular weight (FIG. 2A).

(161) The activity of RlpE was established by incubating the enzyme (0.1 μ M) in reaction buffer (25 mM Tris-HCl pH 8.0, 2.5 mM MgCl₂) with 1 mM of TDP-L-rhamnose and 0.5 mM NB-RLP860. Reactions (200 μ L) were incubated at 30° C. for 4 h. A portion (25 μ L) of the reaction was removed at 15 s, 1 min, 5 min, 20 min, 1 h and 4 h. The reaction was stopped by the addition of two volumes of methanol followed by flash freezing. Quenched reactions were separated by UPLC (Accela™, Thermo Fisher Scientific Mississauga, ON, Canada) and the eluates analyzed by HRESIMS (LTQ Orbitrap Velos; Thermo Fisher Scientific) (positive mode, monitoring m/z 200-2000). Chromatographic separation was achieved with a Hypersil Gold 1.9 μ m C₁₈ 175 Å 50×2.1 mm column (thermo Fisher Scientific) and a linear gradient from 50% H₂O/0.1% FA (solvent A) and 50% acetonitrile (CH₃CN)/0.1% FA (solvent B) to 100% solvent B over 5 min followed by a hold of 100% solvent B for 3 min with a flow rate of 300 μ L/min. Reactions conducted with boiled enzyme showed no conversion of NB-RLP860 to NB-RLP1006. In contrast, enzyme reactions containing intact 6×His-RlpE resulted in the complete conversion of NB-RLP860 to NB-RLP1006 after 4 h (FIG. 2B). This data indicates that RlpE catalyzes the second rhamnosylation step in glycolipopeptide biosynthesis in *V. paradoxus* RKN-096. As genes for the biosynthesis of natural products in bacteria are typically clustered, this finding also provides strong evidence confirming the proposed gene cluster as the locus responsible for glycolipopeptide biosynthesis.

(162) We also explored the ability of purified 6His-RlpE to iteratively add rhamnose units to NB-RLP860 by scanning the HRMS data for masses consistent with glycolipopeptide surfactants containing three rhamnose residues (calc'd [M+H]⁺ 1153.7204), four rhamnose residues (calc'd [M+H]⁺ 1299.7783) and five rhamnose residues (calc'd [M+H]⁺ 1445.8367). Masses consistent with trirhamnosylated and tetrarhamnosylated reaction products were obtained and differed from the expected molecular weights by <1.03 parts per million (ppm) and <0.6 ppm, respectively. Interestingly, two peaks were observed for each mass, suggesting additional rhamnose units are attached at two different positions of the NB-RLP1006 structure. Relative to the production of NB-RLP1006 the tri-rhamnosylated and tetra-rhamnosylated glycolipopeptides constituted 2.99% and 0.14% of the reaction products. No penta-rhamnosylated glycolipopeptides were detected. This data indicates that recombinantly expressed RlpE can be used to generate glycolipopeptide analogs with up to four rhamnose residues. Such modifications may alter the functional properties of the glycolipopeptide. The properties can include but are not limited to wetting, foaming, surfactancy and emulsification.

(163) Elucidation of the biosynthetic pathway for the glycolipopeptide biosurfactants produced by *V. paradoxus* RKNM-096 sets the stage for rational modification of the biosynthetic pathway to generate novel analogues or to increase yields. Analogues may be generated by those skilled in the art via modification of the enzymes responsible for the biosynthesis and incorporation of the lipid, peptide and carbohydrate portions of the molecule. Yields can be increased by those skilled in the art by modification of regulatory genes and or promoters, by overexpressing enzymes that represent rate limiting steps in the biosynthetic pathway or by inactivating enzymes which perform undesirable reactions. Knowledge of the biosynthetic pathway also enables expression in a heterologous host, which may enable yield improvements or the generation of glycolipopeptide analogues.

Example 8: Identification of Related Biosurfactants in Other Bacteria

(164) Sequencing of the *V. paradoxus* RKNM-096 glycolipopeptide and rhamnose biosynthetic gene clusters was performed. Prior to the discovery of the glycolipopeptide series of biosurfactants and the associated biosynthetic gene cluster described herein, it would not have been possible to accurately predict the production of related glycolipopeptide biosurfactants based solely on DNA

sequence analysis. Identification of the glycolipopeptide biosynthetic gene cluster now allows for targeted interrogation of microbial genomes for related gene clusters, which may have the potential to produce novel glycolipopeptide biosurfactants. As *rlpC* encodes a novel rhamnosyltransferase, which glycosylates an acylated dipeptide intermediate characteristic of the glycolipopeptide class of biosurfactants, we used the deduced amino acid sequence of this gene to search available bacterial genomes for homologs. This search identified homologs exhibiting to RlpC from a wide variety of bacteria. We then investigated genomic regions flanking the genes encoding the RlpC homologs for the presence of homologs of the other glycolipopeptide biosynthetic genes. Two examples will be presented to demonstrate the utility of using sequences from the glycolipopeptide gene cluster as probes to discover producers of putatively novel biosurfactants.

(165) A homologous gene cluster was identified in the *Janthinobacterium agaricidamnosum* DSM 9628 genome (GenBank accession no. NZ_HG322949.1) (FIG. 3). *J. agaricidamnosum* is a beta-proteobacterium like *V. paradoxus*, but belongs to a different family. The RlpC homolog in this strain (WP_038493268.1) exhibited 68% identity to RlpC. Scanning the genome around the RlpC homolog identified other homologs of genes present in the glycolipopeptide gene cluster. Directly downstream of the RlpC homolog was an MtbH-like protein (WP_038493269.1) which shared 69% identity with RlpD. Upstream a dimodular NRPS was identified (WP038499875.1), which showed 68% identity to RlpB and contained an identical domain organization ([C-A-PCP].sub.M1-[C-A-PCP-R].sub.M2). Active site analysis (Bachmann and Ravel [2009] *Meth. Enzymol.* 458, 181) indicated that the predicted substrate specificity also matched that of RlpB, with the M1 A-domain specificity code matching that for L-serine and the M2 A-domain specificity code matching that of the M2 A-domain of RlpB, indicating L-leucine is incorporated by M2 (FIG. 3). A C-domain and R-domain were also found at the amino and carboxy-termini of the *J.*

agaricidamnosum NRPS, respectively. This suggests that biosynthesis is initiated by condensation of an acyl intermediate with serine, and terminated by reductive release of an acylated dipeptide, similar to what is predicted for glycolipopeptide biosynthesis in *V. paradoxus* RKNM-096. No homolog to RlpE was found in the *J. agaricidamnosum* DSM 9628 gene cluster, indicating that the product of the cluster likely contains a single rhamnose residue. A gene cluster with a highly similar organization to that in *J. agaricidamnosum* DSM 9628 was also detected in the genome of *V. paradoxus* DSM 21786 (GenBank accession no. NC_022247.1). Collectively, this data suggests that *J. agaricidamnosum* DSM9628 and *V. paradoxus* DSM 21786 possess the ability to produce novel biosurfactants with structures related to those produced by *V. paradoxus* RKNM-096. Based on the bioinformatics analysis presented here, we predict the compound(s) produced by these bacteria would be a N-acylated L-serinyl-L-leucinol dipeptide bearing a single rhamnose residue.

(166) Genome scanning using the RlpC sequence also identified a putative biosurfactant gene cluster in the more distantly related alpha-proteobacterium *Inquilinus limosus* DSM 16000 (Genbank accession no. NZ_AUHM01000002.1) (FIG. 3). The RlpC homolog (WP_026869107.1) in *I. limosus* shared 61% identity with the *V. paradoxus* RKNM-096 protein. Genes encoding a MtbH-like protein (WP_026869104.1) and a NRPS (WP026869105.1) were identified immediately upstream of the RlpC homolog. The MbtH-like protein shared 56% identity with RlpD. The NRPS was a monomodular enzyme with the following domain organization: C-A-T-R (FIG. 3). Active site analysis of the A-domain (Bachmann and Ravel [2009] *Meth. Enzymol.* 458, 181) indicated that L-serine is the likely substrate of this enzyme. The presence of a C-domain at the N-terminus and an R-domain at the C-terminus suggests that the product of the NRPS is an acylated serinol. An RlpE homolog (WP_034850803.1) was also detected in the *I. limosus* gene cluster (41% identity) suggesting that the acylated serinol intermediate may be sequentially glycosylated to yield a product bearing a dirhamnosyl moiety similar to NB-RLP1006. The final product may be exported out of the cell via the action of a MFS exporter (WP_034850806.1) which shares 52% identity with RlpF.

(167) To validate our in silico approach to identifying producers of glycolipopeptide biosurfactants

we obtained *J. agaricidamnorum* DSM 9628 and *V. paradoxus* DSM 21786 from the Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ) culture collection. Each strain was fermented in a variety of culture media to promote production of predicted biosurfactants. Fermentations were extracted twice with an equal volume of EtOAc. The organic layer was evaporated and the resulting concentrated extracts were analyzed by UPLC-PDA-ELSD-HRESIMS as described above for NB-RLP1006 (Example 3). Three prominent peaks eluting at 3.07, 5.05 and 5.51 min were observed in the ELSD and HRESIMS chromatograms of *J. agaricidamnorum* DSM 9628. The peak at 3.07 min (HRESIMS m/z 1182.6217 $[M+H].sup.+$, calcd for C.sub.56H.sub.85N.sub.12O.sub.16, 1181.6201) could be attributed to the known compound jagaracin previously reported from this strain (Graupner et al. [2012] Angew Chem. Int. Ed. Engl. 51:13173). Extraction of the mass spectra for peaks eluting at 5.05 and 5.51 min revealed $[M+H].sup.+$ ions of m/z 833.5741 and m/z 861.6033, respectively. The observed $[M+H].sup.+$ ions showed a -1.5 and 1.0 ppm mass difference from predicted m/z $[M+H].sup.+$ ions for the monorhamnosyl glycolipopeptides NB-RLP832 (m/z 833.5741 $[M+H].sup.+$) and NB-RLP860 (m/z 861.6033 $[M+H].sup.+$), respectively, indicating the expected compounds had been produced by *J. agaricidamnorum* DSM 9628. Similar to NB-RLP978A-C and NB-RLP1020A-C, the mass of NB-RLP832 closely matched that predicted for an analogue of NB-RLP860 lacking two methylene groups. These compounds were purified and their structures elucidated using a combination of 1D and 2D NMR experiments. This analysis unambiguously confirmed that the expected monorhamnosylated biosurfactant had been produced by *J. agaricidamnorum* DSM 9628 (see Example 3).

(168) Identical analysis of *V. paradoxus* DSM 21786 fermentation extracts also revealed the presence of a peak eluting at 5.51 min in the HRESIMS chromatogram. Inspection of the mass spectrum associated with this peak revealed the presence of a $[M+H].sup.+$ ion with a m/z of 861.6104, which differed from the expected mass ($[M+H].sup.+$ m/z 861.6046) by 5.8 ppm. The identical retention time and monoisotopic mass indicated that both *J. agaricidamnorum* DSM 9628 and *V. paradoxus* DSM 21786 produce NB-RLP860.

Claims

1. A purified biosurfactant comprising a hydrophobic lipid component comprising a carboxyl end and a hydroxyl end, wherein the lipid component is covalently linked to (i) a peptide or nonproteinogenic amino acid chain at the carboxyl end of the lipid component and (ii) a carbohydrate moiety at the hydroxyl end of the lipid component via a glycosidic linkage, wherein the carbohydrate moiety includes one, two or three rhamnose moieties, wherein one or more rhamnose hydroxy groups are acetylated if there is one rhamnose moiety and two or more rhamnose hydroxy groups are acetylated if there are two rhamnose moieties.
2. The purified biosurfactant according to claim 1, wherein the peptide chain comprises between 2 and 10 amino acids.
3. The purified biosurfactant according to claim 1, wherein the lipid component comprises between 1 and 6 alkanolic acid moieties.
4. The purified biosurfactant according to claim 1, wherein the lipid component comprises an acyl chain, the acyl chain having a length in a range of between C.sub.4 to C.sub.20.
5. The purified biosurfactant according to claim 1, wherein the peptide or nonproteinogenic amino acid chain comprises a serine-leucine dipeptide.
6. The purified biosurfactant according to claim 1, wherein the lipid component comprises three β -hydroxyalkanoic acid moieties.
7. The purified biosurfactant of claim 4, wherein each acyl chain of the lipid component has a length of C.sub.10.
8. The purified biosurfactant according to claim 1, wherein the carbohydrate moiety includes a

ribose moiety attached to the lipid component via a glycosidic linkage.

9. The purified biosurfactant according to claim 1, wherein the carbohydrate moiety includes two ribose moieties.

10. The purified biosurfactant according to claim 1, wherein the lipid component comprises three β -hydroxyalkanoic acid moieties, the length of each acyl chain of the lipid component is C₁₀, and the carbohydrate moiety includes a ribose moiety attached to the lipid component via a glycosidic linkage.
